

TMX320DM320ZHK

Digital Media Processor

Data Sheet Rev a0.9

(‘04/2/19)

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HISTORY

Revision	Date	Notes
α0.0	25 Jul'03	Initial revision.
α0.1	09 Nov'03	Input pin assignment/SIF timing
α0.2	14 Nov'03	Updated EM_WAIT value during reset at normal mode.
α0.3	20 Nov'03	Updated pin list format and added pin mux information/correct TRACE_CLK and TRACE_SYNC assignment.
α0.4	08 DEC'03	Add DSP_XF.
α0.5	17 DEC'03	Updated MemoryStick Interface AC table. Updated notes and ARM/SDRAM clock on OSC Interface and Clock Timing. Updated McBSP tck .
α0.6	21 JAN'04	Updated McBSP/AMSIF/SDRAM/External Memory IF AC timing
α0.7	23 JAN'04	Updated EMIF/MMC/SD IF/SIF/VLYNQ AC timing.
α0.8	28 JAN'04	Updated External Host interface timing and some names of MS timing spec and Power supply sequence.
α0.9	19 FEB'04	Updated external memory interface timing

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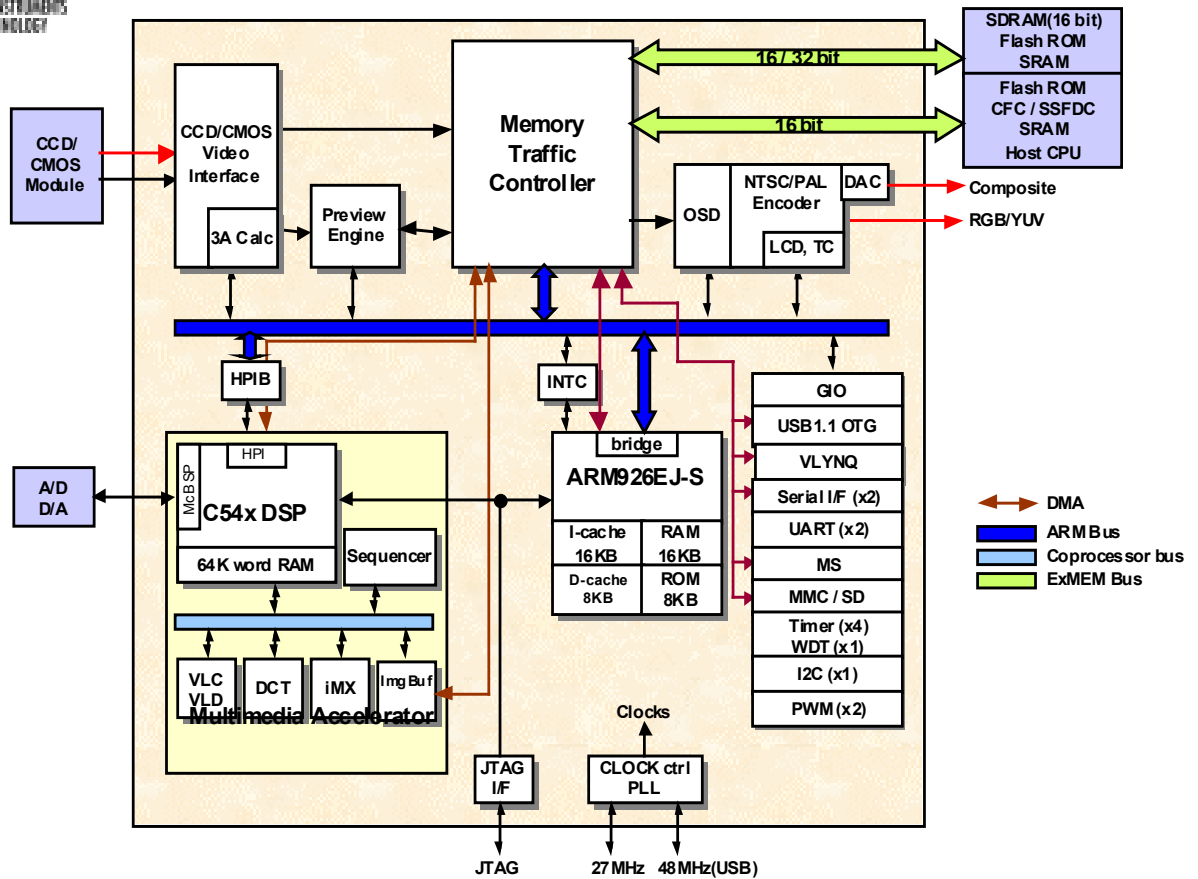
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1. Introduction

1.1 Functional Block Diagram



TMS320DM320 Block Diagram



TMS320DM320



2. Pin Assignments

2.1 Pin List

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Internal Pull up/down
					Normal ²	Ext Host ²		
1	B1	RTCK ⁷	Misc.	I/O	L or X	L or X	Return TCK (Depends on TRST states)	-
2	C2	RXD0 AIMSIF_DI	Communication I/F	I	I	I	UART RXD 0 or AIMSIF Data in	-
3	E5	TXD0 AIMSIF_DO	Communication I/F	O	H	H	UART TXD 0 or AIMSIF Data out	-
4	F7	SIF_DO0	Communication I/F	O	L	L	Serial Data output 0	-
5	C1	SIF_DI0	Communication I/F	I	I	I	Serial Data input 0/	-
6	D3	SIF_EN0	Communication I/F	I/O	I	I	Serial Data Enable 0 / AIMSIF Enable	-
7	F6	DVDD1	Power	-	-	-	I/O VDD	-
8	D2	SIF_CLK0	Communication I/F	I/O	I	I	Serial Clock 0 / AIMSIF clock	-
9	E3	CVDD1	Power	-	-	-	Core VDD	-
10	H8 ¹	VSS1	GND	-	-	-	GND	-
11	D1	FLASH_CE	External memory I/F	O	H	H	Flash ROM Chip Enable	-
12	E2	CF_CE0	External memory I/F	O	H	H	Chip Enable for CF/SSFDC	-
13	F5	CF_CE1	External memory I/F	O	H	H	Chip Enable for CF/Address Latch Enable for SSFDC	-
14	G8	EM_OE	External memory I/F	I/O	H	I(=1) ⁵	Output Enable for FLSH_CE/CE0/CE1	-
15	F3	CF_OE	External memory I/F	O	H	H	Output Enable for CFE1,CFE2; Command Latch Enable for SSFDC	-
16	E1	EM_WE	External memory I/F	I/O	H	I	Write Enable for FLSH_CE/CF_CE0/CF_CE1/EM_CE0/EM_CE1	-
17	G5	CF_IORD	External memory I/F	O	H	H	CFC I/O Read Strobe	-
18	F2	CF_IOWR	External memory I/F	O	H	H	CFC I/O Write Strobe	-
19	F1	CF_IOIS16	External memory I/F	I	I	I	CFC I/O Select	PU
20	G6	EM_WAIT	External memory I/F	I/O	I	I	External Memory Wait	-
21	G3	CF_WAIT	External memory I/F	I	I	I	CFC Wait	PU
22	G7	EM_BEH	External memory I/F	O	H	H	External Memory Byte Enable High	-
23	G2	EM_BEL	External memory I/F	O	H	H	External Memory Byte Enable Low	-
24	G1	EM_CE0	External memory I/F	I/O	H	I(=1) ⁵	External Memory Chip Enable 0	-
25	H5	EM_CE1	External memory I/F	O	H	H	External Memory Chip Enable 1	-
26	H3	CF_RDY	External memory I/F	I	I	I	CFC Ready	PU
27	H6	DVDD2	Power	-	-	-	I/O VDD	-
28	H2	CVDD2	Power	-	-	-	Core VDD	-
29	H9 ¹	VSS2	GND	-	-	-	GND	-
30	H1	BUSWIDTH ⁹	External memory I/F	I	I	I	8/16 bit select for FLSH_CEH	-
31	H7	FLASHSEL ⁹	External memory I/F	I	I	I	Flash ROM Bus Selector	-
32	J5	EXTHOST ¹⁰	External Host I/F	I	I	I	EMIF/EHIF Selector	-
33	J3	EH_BMUX ⁹	External Host I/F	I	I	I	address/data bus multiplex access mode	-
34	J2	ARM_D15	External memory I/F	I/O	I	I	External Memory Data (MSB)	-
35	J1	ARM_D14	External memory I/F	I/O	I	I	External Memory Data	-
36	J6	ARM_D13	External memory I/F	I/O	I	I	External Memory Data	-
37	K1	ARM_D12	External memory I/F	I/O	I	I	External Memory Data	-
38	K2	ARM_D11	External memory I/F	I/O	I	I	External Memory Data	-
39	K3	ARM_D10	External memory I/F	I/O	I	I	External Memory Data	-
40	J7	ARM_D9	External memory I/F	I/O	I	I	External Memory Data	-
41	K5	ARM_D8	External memory I/F	I/O	I	I	External Memory Data	-
42	J8	ARM_D7	External memory I/F	I/O	I	I	External Memory Data	-
43	L1	ARM_D6	External memory I/F	I/O	I	I	External Memory Data	-
44	L2	ARM_D5	External memory I/F	I/O	I	I	External Memory Data	-
45	L3	ARM_D4	External memory I/F	I/O	I	I	External Memory Data	-
46	K6	ARM_D3	External memory I/F	I/O	I	I	External Memory Data	-
47	K7	DVDD3	Power	-	-	-	I/O VDD	-
48	L5	CVDD3	Power	-	-	-	Core VDD	-
49	H10 ¹	VSS3	GND	-	-	-	GND	-
50	M1	ARM_D2	External memory I/F	I/O	I	I	External Memory Data	-
51	M2	ARM_D1	External memory I/F	I/O	I	I	External Memory Data	-
52	L6	ARM_D0	External memory I/F	I/O	I	I	External Memory Data(LSB)	-

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Internal Pull up/down
					Normal ²	Ext Host ²		
53	M3	ARM_A22	External memory I/F	I/O	L	I	External Memory Address(MSB)	-
54	L7	ARM_A21	External memory I/F	I/O	L	I	External Memory Address	-
55	M5	ARM_A20	External memory I/F	I/O	L	I	External Memory Address	-
56	N1	ARM_A19	External memory I/F	I/O	L	I	External Memory Address	-
57	N2	ARM_A18	External memory I/F	I/O	L	I	External Memory Address	-
58	M6	ARM_A17	External memory I/F	I/O	L	I	External Memory Address	-
59	N3	ARM_A16	External memory I/F	I/O	L	I	External Memory Address	-
60	P1	ARM_A15	External memory I/F	I/O	L	I	External Memory Address	-
61	M7	ARM_A14	External memory I/F	I/O	L	I	External Memory Address	-
62	P2	ARM_A13	External memory I/F	I/O	L	I	External Memory Address	-
63	N5	ARM_A12	External memory I/F	I/O	L	I	External Memory Address	-
64	P3	ARM_A11	External memory I/F	I/O	L	I	External Memory Address	-
65	R1	ARM_A10	External memory I/F	I/O	L	I	External Memory Address	-
66	R2	ARM_A9	External memory I/F	I/O	L	I	External Memory Address	-
67	T1	DVDD4	Power	-	-	-	I/O VDD	-
68	M8	CVDD4	Power	-	-	-	Core VDD	-
69	H11 ¹	VSS4	GND	-	-	-	GND	-
70	P5	ARM_A8	External memory I/F	I/O	L	I	External Memory Address	-
71	R3	ARM_A7	External memory I/F	I/O	L	I	External Memory Address	-
72	T2	ARM_A6	External memory I/F	I/O	L	I	External Memory Address	-
73	U2	ARM_A5	External memory I/F	I/O	L	I	External Memory Address	-
74	T3	ARM_A4	External memory I/F	I/O	L	I	External Memory Address	-
75	U1	ARM_A3	External memory I/F	I/O	L	I	External Memory Address	-
76	V2	ARM_A2	External memory I/F	I/O	L	I	External Memory Address	-
77	V1	ARM_A1	External memory I/F	I/O	L	I	External Memory Address	-
78	U3	ARM_A0	External memory I/F	I/O	L	I	External Memory Address(LSB)	-
79	W2	DSP_BDX0	Audio I/F	O	HiZ	HiZ	DSP Serial-port Transmit 0	-
		TRACE_PKT6 ⁶	ARM ETM9			X	ARM Embedded Trace Data	
80	N6	DVDD5	Power	-	-	-	I/O VDD	-
81	V3	DSP_BCLKS0	Audio I/F	I/O	I	I	DSP Serial Clock 0	-
		TRACE_PKT0 ⁶	ARM ETM9			X	ARM Embedded Trace Data	
82	P6	CVDD5	Power	-	-	-	Core VDD	-
83	W3	DSP_BCLKX0	Audio I/F	I/O	I	I	DSP Transmit Clock 0	-
		TRACE_PKT5 ⁶	ARM ETM9			X	ARM Embedded Trace Data	
84	U4	DSP_BFSX0	Audio I/F	I/O	I	I	DSP Frame Sync Pulse Transmit 0	-
		TRACE_PKT4 ⁶	ARM ETM9			X	ARM Embedded Trace Data	
85	V4	DSP_BDR0	Audio I/F	I/O	I	I	DSP Serial Data Receive 0	-
		TRACE_PKT3 ⁶	ARM ETM9			X	ARM Embedded Trace Data	
86	W4	DSP_BCLKR0	Audio I/F	I/O	I	I	DSP Receive Clock 0	-
		TRACE_PKT2 ⁶	ARM ETM9			X	ARM Embedded Trace Data	
87	U5	DSP_BFSR0	Audio I/F	I/O	I	I	DSP Frame Sync Pulse Receive 0	-
		TRACE_PKT1 ⁶	ARM ETM9			X	ARM Embedded Trace Data	
88	V5	SDR_A14	SDRAM I/F	O	L	L	SDRAM Bank 0	-
89	W5	SDR_A13	SDRAM I/F	O	L	L	SDRAM Bank 1	-
90	H12 ¹	VSS5	GND	-	-	-	GND	-
91	R6	SDR_A12	SDRAM I/F	O	L	L	SDRAM Address(MSB)	-
92	U6	SDR_A11	SDRAM I/F	O	L	L	SDRAM Address	-
93	V6	SDR_A10	SDRAM I/F	O	L	L	SDRAM Address	-
94	W6	SDR_A9	SDRAM I/F	O	L	L	SDRAM Address	-
95	R7	SDR_A8	SDRAM I/F	O	L	L	SDRAM Address	-
96	N7	DVDD_SDR1	Power	-	-	-	I/O VDD for SDRAM	-
97	N8	CVDD6	Power	-	-	-	Core VDD	-
98	U7	SDR_A7	SDRAM I/F	O	L	L	SDRAM Address	-
99	V7	SDR_A6	SDRAM I/F	O	L	L	SDRAM Address	-
100	W7	SDR_A5	SDRAM I/F	O	L	L	SDRAM Address	-
101	P7	SDR_A4	SDRAM I/F	O	L	L	SDRAM Address	-
102	R8	SDR_A3	SDRAM I/F	O	L	L	SDRAM Address	-
103	U8	SDR_A2	SDRAM I/F	O	L	L	SDRAM Address	-
104	V8	SDR_A1	SDRAM I/F	O	L	L	SDRAM Address	-
105	P8	SDR_A0	SDRAM I/F	O	L	L	SDRAM Address(LSB)	-
106	W8	SDR_CAS	SDRAM I/F	O	H	H	SDRAM Column Address	-
107	N9	SDR_RAS	SDRAM I/F	O	H	H	SDRAM Row Address	-
108	P9	SDR_WE	SDRAM I/F	O	H	H	SDRAM write enable	-

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Internal Pull up/down
					Normal ²	Ext Host ²		
109	R9	SDR_CS	SDRAM I/F	O	H	H	SDRAM Chip Select	-
110	U9	SDR_CKE	SDRAM I/F	O	H	H	SDRAM Clock Enable	-
111	V9	DVDD_SDR2	Power	-	-	-	I/O VDD for SDRAM	-
112	W9	SDR_CLK	SDRAM I/F	I/O	L	L	SDRAM clock *4	-
113	J9 ¹	VSS6	GND	-	-	-	GND	-
114	M10	CVDD7	Power	-	-	-	Core VDD	-
115	N10	SDR_DQM3	SDRAM I/F	O	H	H	DQMH for SDR_DQ[31:24]	-
116	N11	SDR_DQM2	SDRAM I/F	O	H	H	DQMH for SDR_DQ[23:16]	-
117	U10	SDR_DQM1	SDRAM I/F	O	H	H	DQMH for SDR_DQ[15:8]	-
118	V10	SDR_DQM0	SDRAM I/F	O	H	H	DQMH for SDR_DQ[7:0]	-
119	W10	SDR_D31	SDRAM I/F	I/O	I	I	SDRAM Data (MSB)	PD/R
120	R10	SDR_D30	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
121	P10	SDR_D29	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
122	P11	SDR_D28	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
123	N12	SDR_D27	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
124	U11	SDR_D26	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
125	V11	SDR_D25	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
126	W11	SDR_D24	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
127	R11	SDR_D23	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
128	P12	SDR_D22	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
129	M11	DVDD_SDR3	Power	-	-	-	I/O VDD for SDRAM	-
130	J10 ¹	VSS7	GND	-	-	-	GND	-
131	R12	CVDD8	Power	-	-	-	Core VDD	-
132	U12	SDR_D21	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
133	V12	SDR_D20	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
134	W12	SDR_D19	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
135	N13	SDR_D18	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
136	W13	SDR_D17	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
137	V13	SDR_D16	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
138	U13	SDR_D15	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
139	W14	SDR_D14	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
140	V14	SDR_D13	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
141	P13	SDR_D12	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
142	R13	SDR_D11	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
143	W15	SDR_D10	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
144	U14	SDR_D9	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
145	V15	SDR_D8	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
146	U15	SDR_D7	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
147	M12	DVDD_SDR4	Power	-	-	-	I/O VDD for SDRAM	-
148	J11 ¹	VSS8	GND	-	-	-	GND	-
149	R14	CVDD9	Power	-	-	-	Core VDD	-
150	W16	SDR_D6	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
151	V16	SDR_D5	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
152	U16	SDR_D4	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
153	W17	SDR_D3	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
154	V17	SDR_D2	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
155	W18	SDR_D1	SDRAM I/F	I/O	I	I	SDRAM Data	PD/R
156	U17	SDR_D0	SDRAM I/F	I/O	I	I	SDRAM Data(LSB)	PD/R
157	V18	GIO27 UART_RXD1	General I/O Communication I/F	I/O	I	I	GIO UART Receive data 1	-
158	V19	GIO28 UART_TXD1	General I/O Communication I/F	I/O	I	I	GIO or UART Transmit data 1	-
159	U18	GIO29 SIF_EN1	General I/O Communication I/F	I/O	I	I	GIO or Serial Data Enable 1	-
160	J12 ¹	VSS9	GND	I/O	I	I	GND	-
161	T17	GIO30 I2C_CLK SIF_CLK1	General I/O Control Communication I/F	I/O	I	I	GIO or I2C clock or Serial Clock 1	-
162	P14	DVDD6	Power	-	-	-	I/O VDD	-
163	U19	GIO31 I2C_DATA SIF_DI1	General I/O Control Communication I/F	I/O	I	I	GIO I2C data1 Serial Data input 1	-

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Internal Pull up/down
					Normal ²	Ext Host ²		
164	T18	GIO32 SIF_DO	General I/O Communication I/F	I/O	I	I	GIO Serial Data Output 1	-
165	R17	GIO33 PWM0 CLKOUT1B	General I/O	I/O	I	I	GIO PWM 0 General clock output 1B	-
166	P15	GIO34 PWM1 R2 CLKOUT2B	General I/O	I/O	I	I	GIO PWM 1 RGB666(R2) General clock output 2B	-
167	T19	GIO35 TMRC0	General I/O	I/O	I	I	GIO Camera Timer 0	-
168	R18	GIO36 TMRC1	General I/O	I/O	I	I	GIO Camera Timer 1	-
169	R19	GIO37 MIRQ MRST	General I/O	I/O	I	I	GIO IRQ Monitor Master Reset from WDT	-
170	P17	GIO38 EXT_LINE_ID B2	General I/O	I/O	I	I	GIO NTSC/PAL Encoder RGB666(B2)	-
171	N15	GIO39 FIELD_VENC	General I/O	I/O	I	I	GIO Field (NTSC/PAL/Enc.) (LCD_OE)	-
172	P18	GIO40 C_WEN	General I/O	I/O	I	I	GIO CCD Inj Write Enable	-
173	N14	CVDD10	Power	-	-	-	Core VDD	-
174	P19	SD_CLK		I/O	L	L	SD card clock	-
175	K8 ¹	VSS10	GND	-	-	-	GND	-
176	M13	SD_CMD	Communication I/F	I/O	I	I	SD card Command	-
177	N17	SD_DATA3	Communication I/F	I/O	I	I	SD card Data 3	-
178	M15	SD_DATA2	Communication I/F	I/O	I	I	SD card Data 2	-
179	N18	SD_DATA1	Communication I/F	I/O	I	I	SD card Data 1	-
180	N19	SD_DATA0	Communication I/F	I/O	I	I	SD card Data 0	-
181	M14	DVDD7	Power	-	-	-	I/O VDD	-
182	L13	GIO10 INT10 DSP_BDX1	General I/O	I/O	I	I	GIO External interrupt DSP Serial-port Transmit 1	-
183	M17	GIO11 INT11 DSP_BCLKX1	General I/O	I/O	I	I	GIO External interrupt DSP Transmit Clock 1	-
184	L14	GIO12 INT12 DSP_BFSX1	General I/O	I/O	I	I	GIO External interrupt DSP Frame Sync Pulse Transmit 1	-
185	M18	GIO13 INT13 DSP_BDR1	General I/O	I/O	I	I	GIO External interrupt DSP Serial Data Receive 1	-
186	M19	GIO14 INT14 DSP_BCLKR1	General I/O	I/O	I	I	GIO External interrupt DSP Receive Clock 1	-
187	L15	GIO15 INT15 DSP_BFSR1	General I/O	I/O	I	I	GIO External interrupt DSP Frame Sync Pulse Receive 1	-
188	K13	CVDD11	Power	-	-	-	Core VDD	-
189	K14	GIO16 DSP_BCLKS1 CLKOUT0	General I/O	I/O	I	I	GIO DSP Serial Clock 1 General clock output(12Mhz)	-
190	K9 ¹	VSS11	GND	-	-	-		-
191	K15	USB_VDD	Power	-	-	-	USB VDD	-
192	L17	USB_DM	Communication I/F	I/O	I	I	USB Data Minus	-
193	L18	USB_DP	Communication I/F	I/O	I	I	USB Data Plus	-

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Internal Pull up/down
					Normal ²	Ext Host ²		
194	L19	GIO17 VBUSCHARGE VLYNQ_RXD0	General I/O	I/O	I	I	GIO USB VBUS Charge VLYNQ receive data 0	PD/V
195	K17	GIO18 VBUSEN VLYNQ_RXD1	General I/O	I/O	I	I	GIO USB VBUS power enable VLYNQ receive data 1	PD/V
196	K18	GIO19 VBUSVAL VLYNQ_RXD2	General I/O	I/O	I	I	GIO USB VBUS Valid threshold VLYNQ receive data 2	PD/V
197	K19	GIO20 VBUSSES VLYNQ_RXD3	General I/O	I/O	I	I	GIO USB VBUS session Valid threshold VLYNQ receive data 3	PD/V
198	J19	GIO21 VBUSLO VLYNQ_TXD0	General I/O	I/O	I	I	GIO USB VBUS Session End threshold VLYNQ transmit data 0	-
199	J14	DVDD8	Power	-	-	-	I/O VDD	-
200	J17	GIO22 CID VLYNQ_TXD1	General I/O	I/O	I	I	GIO Connector ID 0:A-type, 1:B-type VLYNQ transmit data 1	-
201	J18	GIO23 PUCON VLYNQ_TXD2	General I/O	I/O	I	I	GIO D+ pull-up control VLYNQ transmit data 2	-
202	J15	GIO24 PDCON VLYNQ_TXD3	General I/O	I/O	I	I	GIO D- pull-down control VLYNQ transmit data 3	-
203	H19	GIO25 IDEN VLYNQ_SCRUN CLKOUT1A	General I/O	I/O	I	I	GIO CID pull-up control VLYNQ clock run General clock output 1A	PU/V
204	K10 ¹	VSS12	GND	-	-	-	GND	-
205	H18	GIO26 VLYNQ_CLOCK CLKOUT2A	General I/O	I/O	I	I	GIO VLYNQ clock General clock out 2A	PU/V
206	J13	CVDD12	Power	-	-	-	Core VDD	-
207	H17	C10	Video Input	I	I	I	C IN signal	PD
208	H14	C11	Video Input	I	I	I	C IN signal	PD
209	G19	C12	Video Input	I	I	I	C IN signal	PD
210	G18	C13	Video Input	I	I	I	C IN signal	PD
211	H15	C14	Video Input	I	I	I	C IN signal	PD
212	G17	C15	Video Input	I	I	I	C IN signal	PD
213	H13	C16 TRACE_CLK ⁶	Video Input	I/O	I	X	C IN signal or ETM SYNC signal	PD
214	G14	C17 TRACE_SYNC ⁶	Video Input	I/O	I	X	C IN signal or ETM clock out signal	PD
215	F19	VD	Video Input	I/O	I	I	V sync	-
216	F18	HD	Video Input	I/O	I	I	H sync	-
217	F17	FIELD_CCD	Video Input	I/O	I	I	Field Indicator	-
218	G15	DVDD9	Power	-	-	-	I/O VDD	-
219	E19	PCLK	Video Input	I	I	I	Pixel clock	-
220	K11 ¹	VSS13	GND	-	-	-	GND	-
221	E18	Y10 TRACE_PSTATE0 ⁶	Video Input	I/O	I	I	Y IN signal or ETM PIPESTATE signal	PD
222	D19	Y11 TRACE_PSTATE1 ⁶	Video Input	I/O	I	L	Y IN signal or ETM PIPESTATE signal	PD
223	D18	Y12 TRACE_PSTATE2 ⁶	Video Input	I/O	I	L	Y IN signal or ETM PIPESTATE signal	PD
224	E17	Y13 TRACE_PKT7 ⁶	Video Input	I/O	I	X	Y IN signal or ETM trace output signal	PD
225	F15	Y14	Video Input	I	I	I	Y IN signal or	PD
226	C19	Y15	Video Input	I	I	I	Y IN signal	PD

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Internal Pull up/down
					Normal ²	Ext Host ²		
227	D17	YI6	Video Input	I	I	I	Y IN signal	PD
228	C18	YI7	Video Input	I	I	I	Y IN signal	PD
229	K12 ¹	VSS14	GND	-	-	-	GND	-
230	B19	M48XO	Misc.	O	-	-	48MHz X'tal output	-
231	B18	M48XI	Misc.	I/O	I	I	48MHz X'tal input	-
232	C17	CVDD13	Power	-	-	-	Core VDD	-
233	F14	DVDD10	Power	-	-	-	I/O power	-
234	A18	SYSCCLK	Misc.	I	I	I	External Clock Input	-
235	B17	MXO	Misc.	O	-	-	27Mhz X'tal output	-
236	A17	MXI	Misc.	I/O	I	I	27Mhz X'tal input	-
237	L8 ¹	VSS15	GND	-	-	-	GND	-
238	C16	BTSEL0	Misc.	I	I	I	ARM Boot Selector 00: boot up from external memory 01: boot up after loading to AIM or by ext host 10: boot up from NAND-flash using ARM(ROM) 11: same function as BTSEL[1:0]=00	-
239	B16	BTSEL1	Misc.	I	I	I		-
240	E14	PLLVD1	Power	-	-	-	VDD for PLL1	-
241	C15	PLLVS1	Power	-	-	-	GND for PLL1	-
242	E13	CVDD14	GND	-	-	-	Core VDD	-
243	A16	DVDD11	Power	-	-	-	I/O VDD	-
244	C14	TEST2	Misc.	I	I	I	Test Input 2 Set to '0' for normal operation	PD
245	B15	PLLVD2	Power	-	-	-	VDD for PLL2	-
246	A15	PLLVS2	GND	-	-	-	GND for PLL2	-
247	F13	CVDD15	Power	-	-	-	Core VDD	-
248	L9 ¹	VSS16	GND	-	-	-	GND	-
249	G13	SCANEN	Misc.	I	I	I	Test Scan Enable	PD
250	C13	TCK	Misc.	I	I	I	JTAG Clock	PD
251	B14	TDI	Misc.	I	I	I	JTAG Data Input	PU
252	A14	TMS	Misc.	I	I	I	JTAG Test Mode Select	PU
253	F12	TRST	Misc.	I	I	I	JTAG Test Reset	PD
254	B13	DVDD12	Power	-	-	-	I/O VDD	-
255	A13	TDO	Misc.	O	HiZ	HiZ	JTAG Data Output	-
256	C12	VDD3V	Power	-	-	-	Digital VDD for DAC	-
257	E12	VSS3V	GND	-	-	-	Digital GND for DAC	-
258	B12	VREFIN	Video DAC	I	I	I	DAC Reference voltage	-
259	A12	BIAS	Video DAC	I	I	I	DAC Bias with Compensation Capacitor	-
260	E11	VDDA	Power	-	-	-	Analog VDD for DAC	-
261	A11	VSSA	GND	-	-	-	Analog GND for DAC	-
262	B11	AOUT	Video DAC	O	O	O	DAC output	-
263	C11	IREF	Video DAC	I	I	I	DAC current control	-
264	L10 ¹	VSS17	GND	-	-	-	GND	-
265	G12	TEST1	Misc.	I	I	I	JTAG connection in the normal mode(TEST2=0)	PD
266	F11	TEST0	Misc.	I	I	I	00: ARM to DSP (for multi CPU debugging) 01: ARM only 10: DSP only 11: ARM and DSP (parallel)	PD
267	A10	CLKSEL	Misc.	I	I	I	Clock Selector 0: MXI 1: SYSCLOCK	PD
268	B10	RSTN	Misc.	I	I	I	System Reset	-
269	C10	PWRDN	Sleep Controller	I	I	I	Power Down Mode Set to '0' to stop MXI clock	PD
270	G11	DVDD13	Power	-	-	-	I/O VDD	-
271	E10	YOUT7	Video Output	I/O	I	I	Y OUT Signal	PD
272	A9	YOUT6	Video Output	I/O	I	I	Y OUT Signal	PD
273	F10	YOUT5	Video Output	I/O	I	I	Y OUT Signal	PD
274	B9	YOUT4	Video Output	I/O	I	I	Y OUT Signal	PD
275	C9	YOUT3	Video Output	I/O	I	I	Y OUT Signal	PD
276	G10	YOUT2	Video Output	I/O	I	I	Y OUT Signal	PD
277	E9	YOUT1	Video Output	I/O	I	I	Y OUT Signal	PD
278	A8	YOUT0	Video Output	I/O	I	I	Y OUT Signal	PD
279	L11 ¹	VSS18	GND	-	-	-	GND	-
280	B8	VCLK	Video Output	O	HiZ	HiZ	Video Out Clock	-
281	F9	CVDD16	Power	-	-	-	Core VDD	-

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Internal Pull up/down
					Normal ²	Ext Host ²		
282	C8	HSYNC	Video Output	I/O	I	I	H-Sync for RGB output	PD
283	G9	VSYNC	Video Output	I/O	I	I	V-Sync/Composite-Sync output	PD
284	A7	COUT7	Video Output	I/O	I	I	C OUT Signal	PD
285	B7	COUT6	Video Output	I/O	I	I	C OUT Signal	PD
286	C7	COUT5	Video Output	I/O	I	I	C OUT Signal	PD
287	E8	COUT4	Video Output	I/O	I	I	C OUT Signal	PD
288	A6	COUT3	Video Output	I/O	I	I	C OUT Signal	PD
289	B6	COUT2	Video Output	I/O	I	I	C OUT Signal	PD
290	F8	COUT1	Video Output	I/O	I	I	C OUT Signal	PD
291	C6	COUT0	Video Output	I/O	I	I	C OUT Signal	PD
292	E7	DVDD14	Power	-	-	-	I/O VDD	
293	L12 ¹	VSS19	GND	-	-	-	GND	
294	A5	GIO0 INT0	General IO(Wake up)	I/O	I	I	GIO (wake up) External interrupt	
295	B5	GIO1 INT1	General IO	I/O	I	I	GIO External interrupt	
296	A4	GIO2 INT2	General IO	I/O	I	I	GIO External interrupt	
297	C5	GIO3 INT3	General IO	I/O	I	I	GIO External interrupt	
298	B4	GIO4 INT4	General IO	I/O	I	I	GIO External interrupt	
299	A3	GIO5 INT5	General IO	I/O	I	I	GIO External interrupt	
300	E6	GIO6 INT6	General IO	I/O	I	I	GIO External interrupt	
301	C4	GIO7 INT7	General IO	I/O	I	I	GIO External interrupt	
302	B3	CARD_DET GIO8 INT8	CARD_DET / General IO	I/O	I	I	CARD detector/GIO External interrupt	
303	B2	GIO9 INT9 DSP_XF	General IO	I/O	I	I	GIO External interrupt XF output from DSP	
304	A2	EMU0	Misc	I/O	I	I	Emulator Interrupt0	PU
305	C3	EMU1	Misc	I/O	I	I	Emulator Interrupt1	PU
306	M9	CVDD17	Power	-	-	-	Core VDD	

1	Following pins are connected to common GND: H8,H9,H10,H11,H12,J9,J10,J11,J12,K8,K9,K10,K11,K12,L8,L9,L10,L11,L12
2	EXTHOST pin can select either Normal mode, External Host CPU mode
3	PD/R Pulled down during reset PD/V Pulled down when VLYNQ IF mode, not Pulled down when GIO (default). PU/V Pulled up when VLYNQ IF mode, not Pulled up when GIO (default).
4	The use of SDR_CLK and RTCK as input is prohibited.
5	When in External Host mode, EM_OE or EM_CE0 should be input as '1'
6	ETM pins(TRACE_xxx) are selected by EH_BMUX=1 on normal mode
7	I/O direction for RTCK state depends on TRST status - When TRST=0, RTCK = L (output) - When TRST=1, Divided clock of TCK will be output
8	See External Host information
9	See Memory Traffic Controller (EMIF/SDRC) information
10	GIO pin muxing controlled by FSEL registers in GIO module
11	See Video Encoder information for I/O options

Digital Video Out Options

Pin Name	YCC16	YCC8	RGB666	RGB8	EPSON	CASIO	UDISP	STN
YOUT7	Y7	Data7	R7	Data7	Data5	Data5	Data7	Data7
YOUT6	Y6	Data6	R6	Data6	Data4	Data4	Data6	Data6
YOUT5	Y5	Data5	R5	Data5	Data3	Data3	Data5	Data5
YOUT4	Y4	Data4	R4	Data4	Data2	Data2	Data4	Data4
YOUT3	Y3	Data3	R3	Data3	Data1	Data1	Data3	Data3
YOUT2	Y2	Data2	G7	Data2	Data0	Data0	Data2	Data2
YOUT1	Y1	Data1	G6	Data1	GCP	STBYB	Data1	Data1
YOUT0	Y0	Data0	G5	Data0	LP	RIT	Data0	Data0
COUT7	C7	LCD_AC	G4	LCD_AC	RES	CP	LCD_AC	LCD_AC
COUT6	C6	LCD_OE	G3	LCD_OE	XINH	STH	DVALID	LCD_OE
COUT5	C5	BRIGHT	G2	BRIGHT	YSCL	STB	BRIGHT	BRIGHT
COUT4	C4	PWM	B7	PWM	YSCLD	GPCK	PWM	PWM
COUT3	C3	CSYNC	B6	CSYNC	FRYS	GRES	UDISP_FS	Data11
COUT2	C2		B5		FRYP	POL		Data10
COUT1	C1		B4		FRX			Data9
COUT0	C0		B3		DY	GSRT		Data8
GIO34	-	-	R2	-	-	-	-	-
GIO38	-	-	B2	-	-	-	-	-
GIO39	LCD_OE / FIELD	LCD_OE / FIELD	LCD_OE / FIELD	LCD_OE / FIELD	LCD_OE / FIELD	LCD_OE / FIELD	LCD_OE / FIELD	LCD_OE / FIELD

3. Parameter Information

3.1 Absolute maximum ratings

PARAMETER	MIN	MAX	UNIT
Supply voltage range for core, CVDD	-0.5	1.836	V
Supply voltage range for I/O, DVDD, DVDD_SDR	-0.5	4.0	V
Supply voltage range for PLL, PLLVDD	-0.5	1.836	V
Supply voltage for D/A, VDD3V	-0.5	4.0	V
Supply voltage for D/A, VDDA	-0.5	4.0	V
Supply voltage for USB, USBVDD	-0.5	4.0	V
Input voltage range for input, VI	-0.5	DVDD+0.5 CVDD+0.5 USBVDD+0.5 DVDD_SDR+0.5	V
Output voltage range for output, VO	-0.5	DVDD+0.5 CVDD+0.5 USBVDD+0.5 DVDD_SDR+0.5	V
Clamp current for an input or output	-20	+20	mA
Storage temperature range, Tstg	-65	150	°C

1. Functional operation of the device is not implied at any conditions beyond those listed under recommended operating conditions.

2. Additional absolute maximum ratings for specific I/O functions are listed throughout this document.

3. Conditions in the field that exceed any of the absolute maximum ratings for a particular macro can cause permanent damage to the device.

4. Device reliability may be affected by exposure to absolute-maximum-rated conditions for extended periods.

5. Clamp current flows from an input or output pad to a supply rail through a clamp circuit or an intrinsic diode. Positive current results from an applied input or output voltage that is more than 0.5 V higher (more positive) than the supply voltage, CVDD/PLLVDD/DVDD/USBVDD/DVDD_SDR for dual-supply macros. Negative current results from an applied voltage that is more than 0.5 V less (more negative) than the VSS voltage.

3.2 Recommended operating conditions (DVDD/DVDD_SDR/USBVDD > CVDD/PLLVDD)

PARAMETER	MIN	NOM	MAX	UNIT
CVDD	Supply voltage, Core	1.50	1.60	V
PLLVDD	Supply voltage, PLL	1.50	1.60	V
DVDD	Supply voltage, I/O	3.3	3.6	V
	Supply voltage for SDRAM(Normal SDRAM)		3.6	V
DVDD_SDR	Supply voltage for SDRAM(3.0V Mobile SDRAM)		3.3	V
	Supply voltage for SDRAM(2.5V Mobile SDRAM)		2.7	V
DGND, CGND, PLLGND	Supply voltage, GROUND	0		V
VI(OSC)	Input voltage, OSC (*1)	0	CVDD	V
VI(SDRAM)	Input voltage, SDRAM(*2)	0	DVDD_SDR	V
VI(Other)	Input voltage, Other pins (*3)	0	DVDD	V
VO(OSC)	Output voltage, OSC (*1)	0	CVDD	V
VO(SDRAM)	Output voltage, SDRAM(*2)	0	DVDD_SDR	V
VO(Other)	Output voltage, Other pins (*3)	0	DVDD	V
VIH(OSC)	High-level dc input voltage, OSC (*1)	0.7CVDD		V
VIH(SDRAM)	High-level dc input voltage, SDRAM(*2)	0.7DVDD_SDR		V
VIH(Other)	High-level dc input voltage, Other pins (*3)	0.7DVDD		V
VIL(OSC)	Low-level dc input voltage, OSC (*1)	0	0.3CVDD	V
VIL(SDRAM)	Low-level dc input voltage, SDRAM(*1)	0	0.3DVDD_SDR	V
VIL(Other)	Low-level dc input voltage, Other pins (*2)	0	0.3DVDD	V
V _{hys}	Hysteresis voltage, Pins : (*3)	0.13DVDD		V
I _i	Input current, VI=0 to MAX	-1	+1	μA
T _c	Operating case temperature range	-20	95	°C

*1) OSC : MXI/O, M48XI/O

*2) Other pins : Other than *1) pins

*3) GIO0-15,RSTN,PWRDN,TMS,TRST,TEST0-4,SCANEN,BTSEL0/1,CLKSEL,EMU0/1

Preliminary

3.3 Recommended operating conditions for USB (*1)

	PARAMETER	MIN	NOM	MAX	UNIT
USBVDD	Supply voltage	3.00	3.30	3.60	V
VI (Single-ended)	Input voltage, (*2)	0		USBVDD	V
VI (Differential)	Input voltage, (*3)	0.80		2.50	V
VIH (Single-ended)	High-level dc input voltage, (*2)	2.00			V
VIL (Single-ended)	Low-level dc input voltage, (*2)			0.8	V
VID	Differential input voltage, (*3)	200			mV
VO	Output voltage	0		USBVDD	V
RS	Series resistance (5% tolerance)		27		Ω

*1) Compatible with USB(Universal Serial Bus Interface) specifications, Rev1.1, Sep23, 1998

*2) Single-ended input buffers with hysteresis. Compatible with TTL signal levels

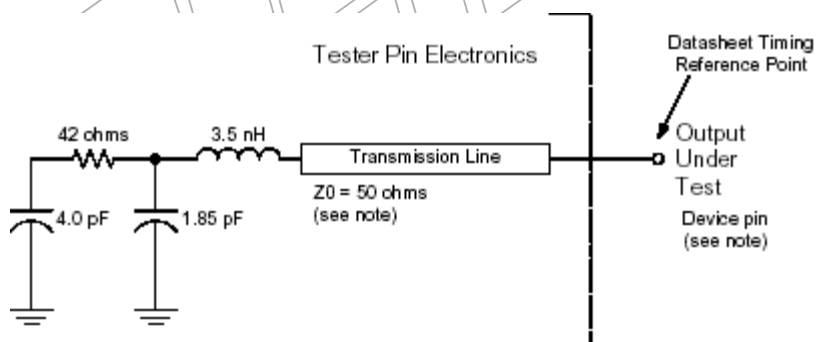
*3) Differential input buffers

3.4 Recommended operating conditions for D/A Converter

	PARAMETER	MIN	NOM	MAX	UNIT
VDD3V	Supply voltage, (digital part)	3.0	3.3	3.6	V
VDDA	Supply voltage, (analog part)	3.0	3.3	3.6	V
VSS3V	Supply voltage, (digital part)		0		V
VSSA	Supply voltage, (analog part)		0		V
VO (*1)	Output voltage	VSSA		Vref	V
Vref	Reference voltage range	1		1.3	V
Riref	Reference register		1.6		kΩ
Raout	Output register		200		Ω
Cbias	Compensation capacitor BIAS to VSSA		0.1		μF

*1) VO range is kept while the ratio (Riref:Raout) have 8:1 relationship.

3.5 Parameter measurement information



Note: The datasheet provides timing at the device pin. For output timing analysis, users must take the tester pin electronics and its transmission line effects into account. Users can use a transmission line with a delay of 2 ns or longer to produce the desired transmission line effect. The transmission line is intended as a load only. Users do not need to add or subtract the transmission line delay (2 ns or longer) from the datasheet timing.

Input requirements in this datasheet are tested with an input slew rate of < 4V/ns at the device pin

Figure 1. Test Load circuit for AC timing measurement

3.6 Electrical characteristics over recommended operating conditions

	PARAMETER	CONDITIONS	MIN	MAX	UNI
VOH(OSC)	High-level output voltage, OSC(*1)	IOH=Rated	0.8CVDD		V
VOH(USB)	High-level output voltage, USB(*2)	IOH=-18.3mA	USB_VDD-0.5		V
VOH(SDR)	High-level output voltage, SDRAM I/F(*3)	IOH=4mA(*5)	0.8DVDD_SDR		V
	High-level output voltage, SDRAM I/F(*3)	IOH=2mA(*6)	2.2		V
	High-level output voltage, SDRAM I/F(*3)	IOH=1mA(*7)	2.2		V
	High-level output voltage, SDR_CLK	IOH=8mA(*5)	0.8DVDD_SDR		V
VOH(SDR_CLK)	High-level output voltage, SDR_CLK	IOH=4mA(*6)	2.2		V
	High-level output voltage, SDR_CLK	IOH=2mA(*7)	2.2		V
VOH(GIO26)	High-level output voltage, GIO26	IOH=8mA	0.8DVDD		V
VOH(other)	High-level output voltage, Other(*4)	IOH=4mA	0.8DVDD		V
VOL(OSC)	Low-level output voltage, OSC(*1)	IOL=Rated		0.22CVDD	V
VOL(USB)	Low-level output voltage, USB(*2)	IOL=18.3mA		0.28	V
VOL(SDR)	Low-level output voltage, SDRAM I/F(*3)	IOL=4mA(*5)		0.22DVDD_SDR	V
	Low-level output voltage, SDRAM I/F(*3)	IOL=2mA(*6)		0.5	V
	Low-level output voltage, SDRAM I/F(*3)	IOL=1mA(*7)		0.2	V
	Low-level output voltage, SDR_CLK	IOL=8mA(*5)		0.22DVDD_SDR	V
VOL(SDR_CLK)	Low-level output voltage, SDR_CLK	IOL=4mA(*6)		0.5	V
	Low-level output voltage, SDR_CLK	IOL=2mA(*7)		0.2	V
VOL(GIO26)	Low-level output voltage, GIO26	IOL=8mA		0.22DVDD	V
VOL(other)	Low-level output voltage, Other(*4)	IOL=4mA		0.22DVDD	V
IIH	Input Leakage	VI = VI max	-1	+1	μA
IIL	Input Leakage	VI = VI max	-1	+1	μA
Ioz	Output High-impedance Leakage		-20	+20	μA
Ci	Input capacitance			10	pF
Co	Output capacitance			10	pF

*1) OSC : MXI/O, M48XI/O

*2) USB : USB_DP/DM

*3) SDRAM I/F; SDRAM pins except SDR_CLK

*4) Other pins : Other than SDR_CLK, GIO26, *1) , *2) and *3) pins

*5) 3.3V Normal SDRAM

*6) 3.0V Mobile SDRAM

*7) 2.5V Mobile SDRAM

4. Timing Diagrams

4.1 OSC Interface and Clock Timing

	PARAMETER	MIN	NOM	MAX	UNIT
tc(M48XI)	Cycle time, M48XI (*1)		48		MHz
tc(MXI)	Cycle time, MXI (*2)		27		MHz
tc(SYSCLK)	Cycle time, external SYSCLK (*6)	0 (PLL bypass) 13.5(PLL input *8)		40	MHz
tc(ARM)	Cycle time, internal ARM clock (*4)			162 or 175.5 *7	MHz
tc(SDR)	Cycle time, Normal SDRAM clock			108 or 117 *7	MHz
	Cycle time, 3.0V mobile SDRAM clock			108 or 117 *7	MHz
	Cycle time, 2.5V mobile SDRAM clock			90	MHz
tc(DSP)	Cycle time, internal DSP clock			99	MHz
tc(iMX)	Cycle time, internal iMX clock			175.5	MHz
tw(RESET)	Pulse duration, RESET low (*3)	10cycle(tc(ARM))			ns
Td(TP)	PLL lock-in period from clock in	500 (*5)			us

*1) For USB and I2C.

*2) For Video output.

*3) RESET need 10cycle (tc(ARM)) after Cycle stable.

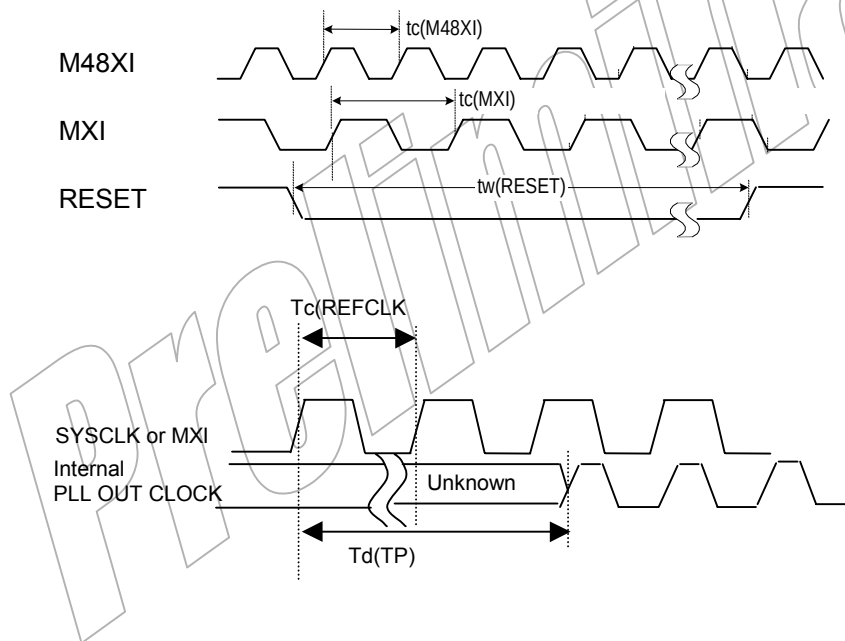
*4) Each cycle time of internal module is controlled by the register on CLKC module. (Please refer "DM320 register manual".)

*5) Under condition the reference clock frequency to PLL = 27MHz.

*6) Duty=50+/-5%

*7) The spec will be fixed after the characterization

*8) PLL reference min clock is 13.5Mhz on PLL spec.



4.2 Audio Interface Timing

timing requirements for McBSP [†]

PARAMETER			MIN	MAX	UNIT
tc(BCKRX)	Cycle time, BCLKR/X	BCLKR/X ext	25		ns
th(BCKRL-BFRH)	Hold time, external BFSR high after BCLKR low	BCLKR int	0		ns
		BCLKR ext	4		
th(BCKRL-BDRV)	Hold time, BDR valid after BCLKR low	BCLKR int	0		ns
		BCLKR ext	6		
th(BCKXL-BFXH)	Hold time, external BFSX high after BCLKX low	BCLKX int	0		ns
		BCLKX ext	10		
tsu(BFRH-BCKRL)	Setup time, external BFSR high before BCLKR low	BCLKR int	20		ns
		BCLKR ext	5		
tsu(BDRV-BCKRL)	Setup time, BDR valid before BCLKR low	BCLKR int	20		ns
		BCLKR ext	4		
tsu(BFXH-BCKXL)	Setup time, external BFSX high before BCLKX low	BCLKX int	20		ns
		BCLKX ext	5		

[†]Polarity bits CLKRP = CLKXP = FSRP = FSXP = 0. If the polarity of any of the signals is inverted, then the timing references of that signal are also inverted.

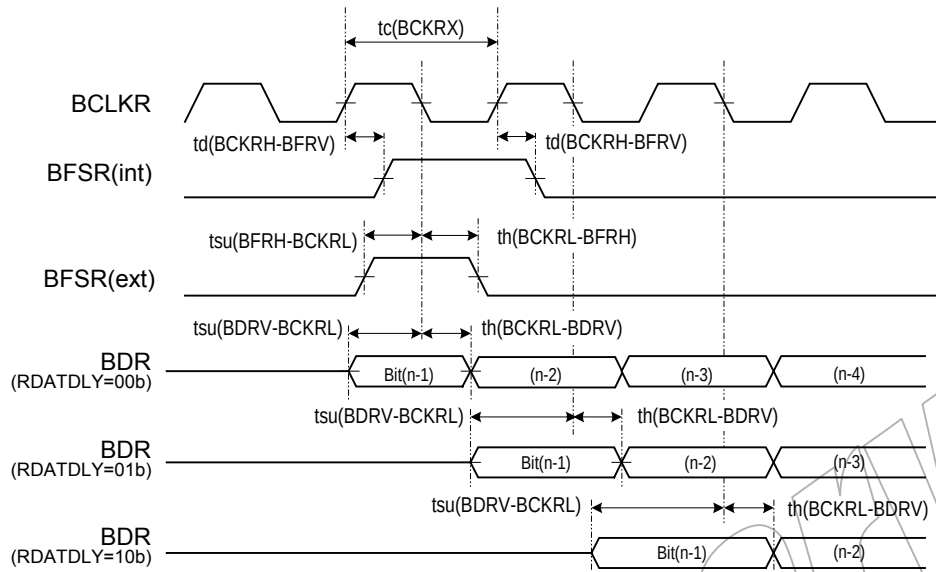
switching characteristics for McBSP [†]

PARAMETER			MIN	MAX	UNIT
tc(CLKS)	Cycle time		25		ns
tc(BCKRX)	Cycle time, BCLKR/X	BCLKR/X int	25		ns
td(BCKRH-BFRV)	Delay time, BCLKR high to internal BFSR valid	BCLKR int	-5	4	ns
td(BCKXH-BFXV)	Delay time, BCLKX high to internal BFSX valid	BCLKX int	-3	8	ns
		BCLKX ext	3	25	
tdis(BCKXH-BDXHZ)	Disable time, BCLKX high to BDX high impedance following last data bit	BCLKX int	-6	5[‡]	ns
		BCLKX ext	0	20[‡]	
td(BCKXH-BDXV)	Delay time, BCLKX high to BDX valid. This applies to all bits except the first bit transmitted.	BCLKXi nt	-3	12	ns
		BCLKX ext	3	25	
	Delay time, BCLKX high to BDX valid. Only applies to first bit transmitted when in Data Delay 1 or 2 (XDATDLY=01b or 10b) modes	BCLKX int		12	ns
		BCLKX ext		25	
te(BCKXH-BDX)	Enable time, BCLKX high to BDX driven. Only applies to first bit transmitted when in Data Delay 1 or 2 (XDATDLY = 01b or 10b) modes.	BCLKX int	-3[‡]		ns
		BCLKX ext	2[‡]		
td(BFXH-BDXV)	Delay time, BFSX high to BDX valid. Only applies to first bit transmitted when in Data Delay 0 (XDATDLY = 00b) mode.	BFSX int		8	ns
		BFSX ext		24	
te(BFXH-BDX)	Enable time, BFSX high to BDX driven. Only applies to first bit transmitted when in Data Delay 0 (XDATDLY = 00b) modes.	BFSX int	-11		ns
		BFSX ext	3		
td(CLKS-BCLKRX)	Delay time,CLKS high to internal BCLKR/X	BCLKR/X	1	24	ns

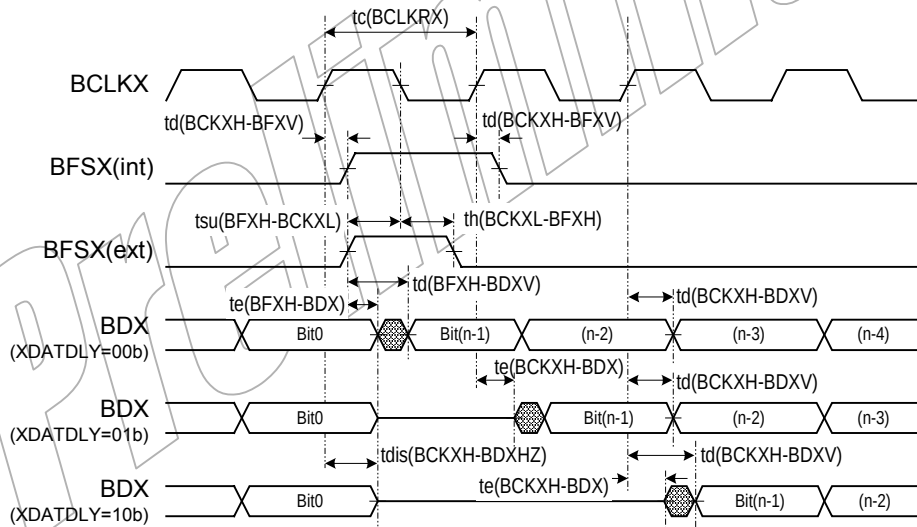
[†]Polarity bits CLKRP = CLKXP = FSRP = FSXP = 0. If the polarity of any of the signals is inverted, then the timing references of that signal are also inverted.

[‡] Value assured by design but not tested.

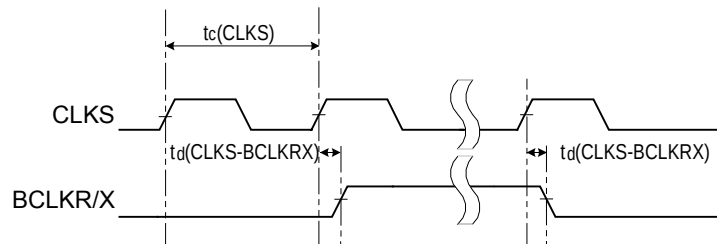
McBSP receive timings



McBSP transmit timings



CLKS Timings



4.3 CCD and Video Interface Timing

CCD interface timing of following parameters

PARAMETER		MIN	NOM	MAX	UNIT
tCLK	PCLK			100	MHz
tS	CI setup time	3			ns
tS	COUT setup time	3			ns
tS	YI setup time	3			ns
tS	YOUT setup time	3			ns
tS	HD setup time	3			ns
tS	VD setup time	3			ns
tS	FIELD_CCD setup time	3			ns
tS	GIO40(C_WEN)	3			ns
TH	CI hold time	2			ns
tH	COUT hold time	2			ns
tH	YI hold time	2			ns
tH	YOUT hold time	2			ns
tH	HD hold time	2			ns
tH	VD hold time	2			ns
tH	FIELD_CCD hold time	2			ns
tH	GIO40(C_WEN)	2			ns
tPD	COUT delay time	0.5		15.5	ns
tPD	YOUT delay time	0.5		15.5	ns
tPD	HD delay time	0.5		15.5	ns
tPD	VD delay time	0.5		15.5	ns
tPD	HSYNC delay time	0.5		15.5	ns
tPD	VSNC delay time	0.5		15.5	ns
tPD	FIELD_CCD delay time	0.5		15.5	ns

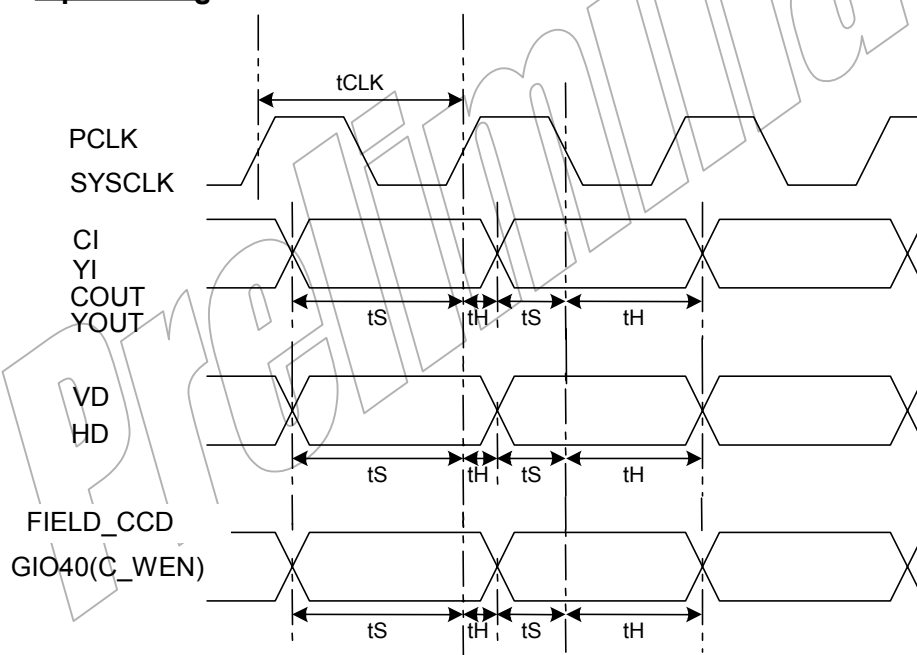
VEDIO ENCODER/LCD IF timing of following parameters.

PARAMETER		MIN	NOM	MAX	UNIT
tCLK	SYSCCLK/PCLK			54	MHz
tCD	VCLK	2.5		15.5	ns
tS	CI setup time	3			ns
tS	COUT setup time	3			ns
tS	YI setup time	3			ns
tS	YOUT setup time	3			ns
tS	HSYNC setup time	3			ns
tS	VSNC setup time	3			ns
tH	CI hold time	2			ns
tH	COUT hold time	2			ns
tH	YI hold time	2			ns
tH	YOUT hold time	2			ns
tH	HSYNC hold time	2			ns
tH	VSNC hold time	2			ns
tPD	COUT delay time	0.5		15.5	ns
tPD	YOUT delay time	0.5		15.5	ns
tPD	GIO39(FIELD_VENC) delay time	0.5		15.5	ns
tPD	GIO34(R2), GIO37(B2) delay time	0.5		15.5	ns
tPD	HSYNC delay time	0.5		15.5	ns
tPD	VSNC delay time	0.5		15.5	ns
tPD	COUT delay time from VCLK	-3		5	ns
tPD	YOUT delay time from VCLK	-3		5	ns
tPD	GIO39(FIELD_VENC) delay time from VCLK	-3		5	ns
tPD	GIO34(R2), GIO37(B2) delay time from VCLK	-3		5	ns
tPD	HSYNC delay time from VCLK	-3		5	ns
tPD	VSNC delay time from VCLK	-3		5	ns

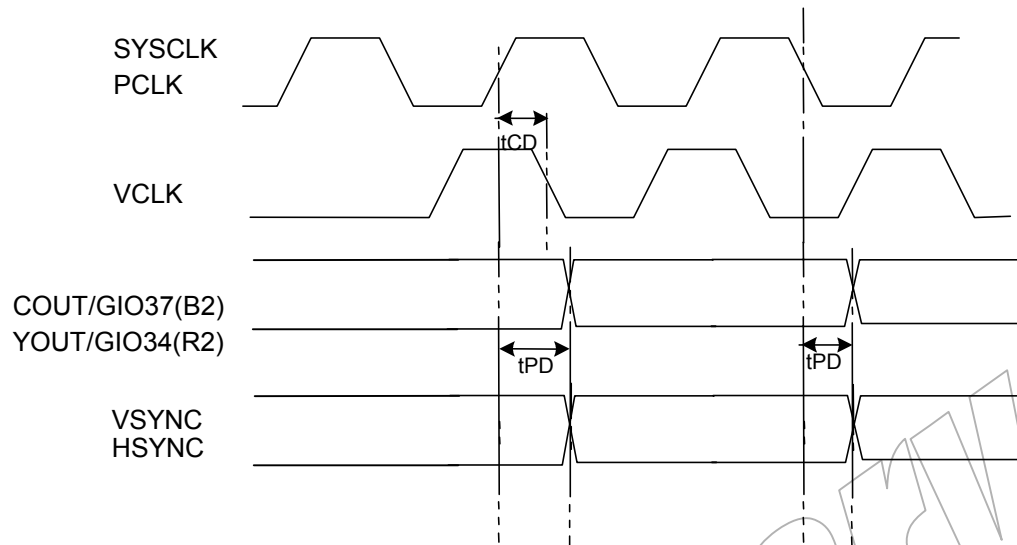
Note) tCLK : Clock cycle time
tCD : Output clock delay time
tS : Data setup time
tH : Data hold time
tPD : Output delay time

Video I/F Pin function Table (LCD Output)

Pin Name	Standard	EPSON	CASIO	Micro Display	STN LCD	RGB666
HSYNC	HSYNC	HSYNC	HSYNC		CL1	HSYNC
VSYNC	VSYNC	VSYNC	VSYNC	FRAME_START	FLM	VSYNC
VCLK	DCLK	DCLK	DCLK	CLK	CL2	DCLK
YOUT7	Data7	Data5	Data5	VIDEO_IN7	DATA7	R7
YOUT6	Data6	Data4	Data4	VIDEO_IN6	DATA6	R6
YOUT5	Data5	Data3	Data3	VIDEO_IN5	DATA5	R5
YOUT4	Data4	Data2	Data2	VIDEO_IN4	DATA4	R4
YOUT3	Data3	Data1	Data1	VIDEO_IN3	DATA3	R3
YOUT2	Data2	Data0	Data0	VIDEO_IN2	DATA2	G7
YOUT1	Data1	GCP	STBYB	VIDEO_IN1	DATA1	G6
YOUT0	Data0	LP	RIT	VIDEO_IN0	DATA0	G5
COUT7	LCD_AC	RES	CP	Reserved	Reserved	G4
COUT6	LCD_OE	XINH	STH	DATA_VALID	Reserved	G3
COUT5	BRIGHT	YSCL	STB	Reserved	Reserved	G2
COUT4	COMP_SYNC	YSCLD	GPCK	DISP_CLK	Reserved	B7
COUT3	PWM_OUT	FRYS	GRES	Reserved	Reserved	B6
COUT2	reserved	FRYP	POL	reserved	Reserved	B5
COUT1	reserved	FRX	reserved	BALANCE	Reserved	B4
COUT0	reserved	DY	GSRT	FULL	Reserved	B3
GIO34	Reserved	Reserved	Reserved	Reserved	Reserved	R2
GIO37	Reserved	Reserved	Reserved	Reserved	Reserved	B2
GIO38	Reserved	Reserved	Reserved	Reserved	Reserved	LCD_OE

Input Timing

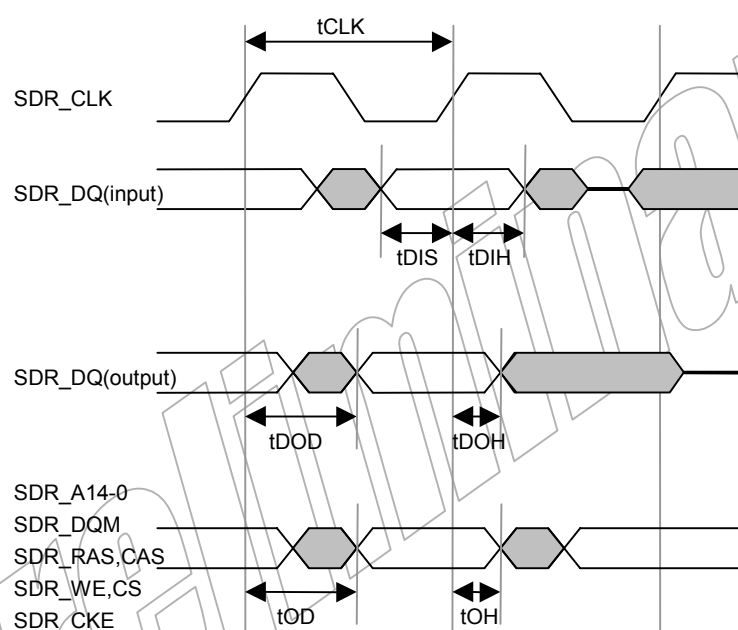
Output Timing



4.4 SDRAM Interface Timing

PARAMETER	MIN	NOM	MAX	UNIT
tCLK	Cycle time, SDR_CLK	9.26	-	ns
tDIS	Input Data setup time	1.8(2.0)		ns
tDIH	Input Data hold time	0.0(0.0)		ns
tDOD	Output Data delay time		5.5(6.2)	ns
tDOH	Output Data hold time	2.0(2.2)		ns
tOD	Output Signal delay time		5.5(6.2)	ns
tOH	Output Signal hold time	2.0(2.2)		ns

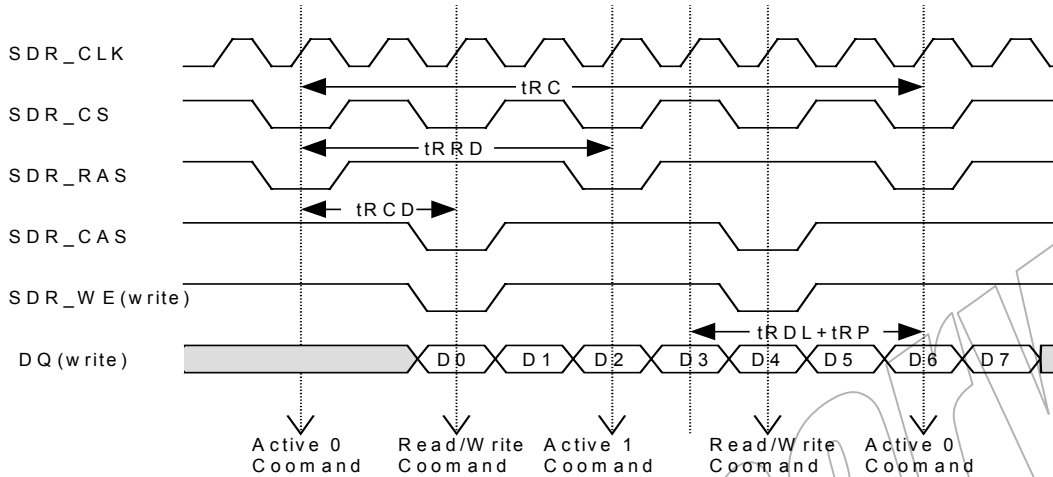
Note. Above () timing data is for SDRAM(2.5V Mobile SDRAM).



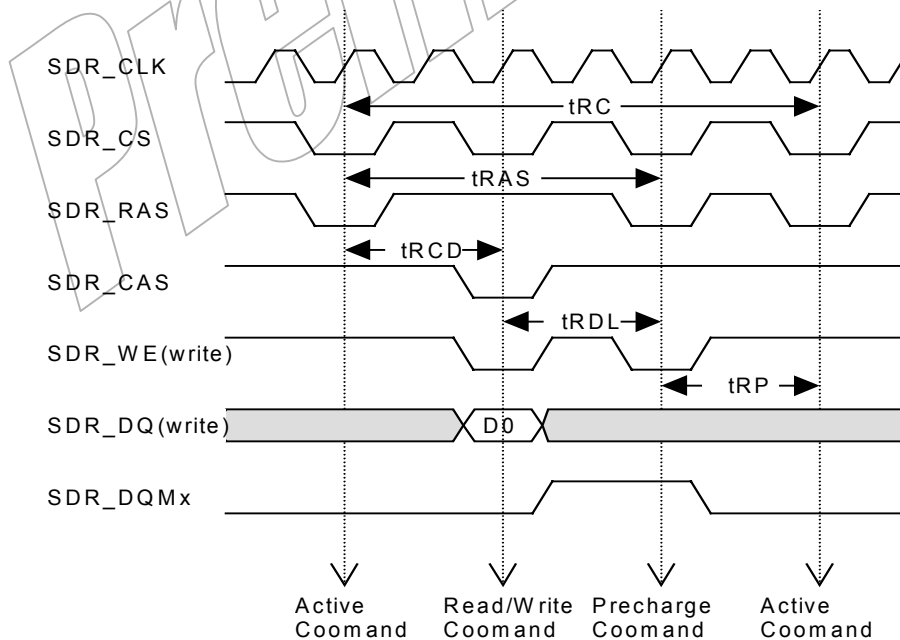
Burst Read/Write with Auto-Precharge (BL = 4)

PARAMETER	CYCLE	UNIT
t _{RC}	8(9)	CLK
t _{RCD}	2	CLK
t _{RRD}	4	CLK
t _{RDL} +t _{RP}	3(4)	CLK

(*): TRDL = 1 (MODESET register)

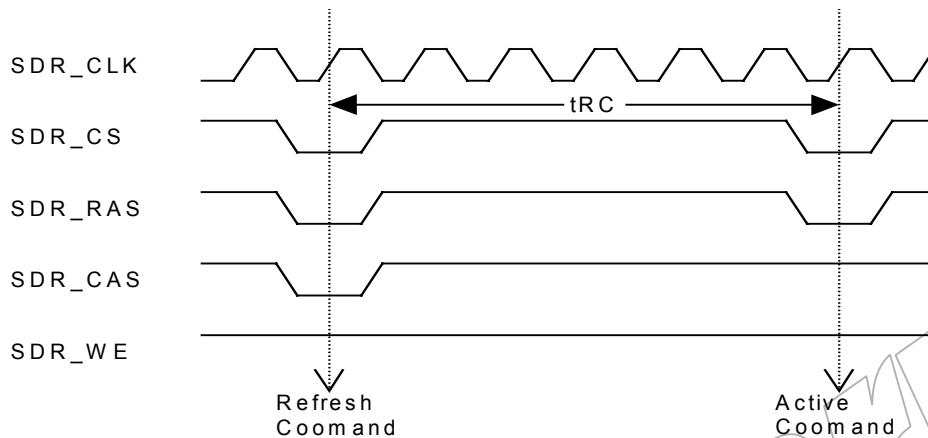
**Burst Interrupt by Precharge (BL = 4)**

PARAMETER	CYCLE	UNIT
t _{RC}	7	CLK
t _{RCD}	2	CLK
t _{RAS}	5	CLK
t _{RDL}	3	CLK
t _{RP}	2	CLK



Auto Refresh

	PARAMETER	CYCLE	UNIT
tRC	Row cycle time	7	CLK

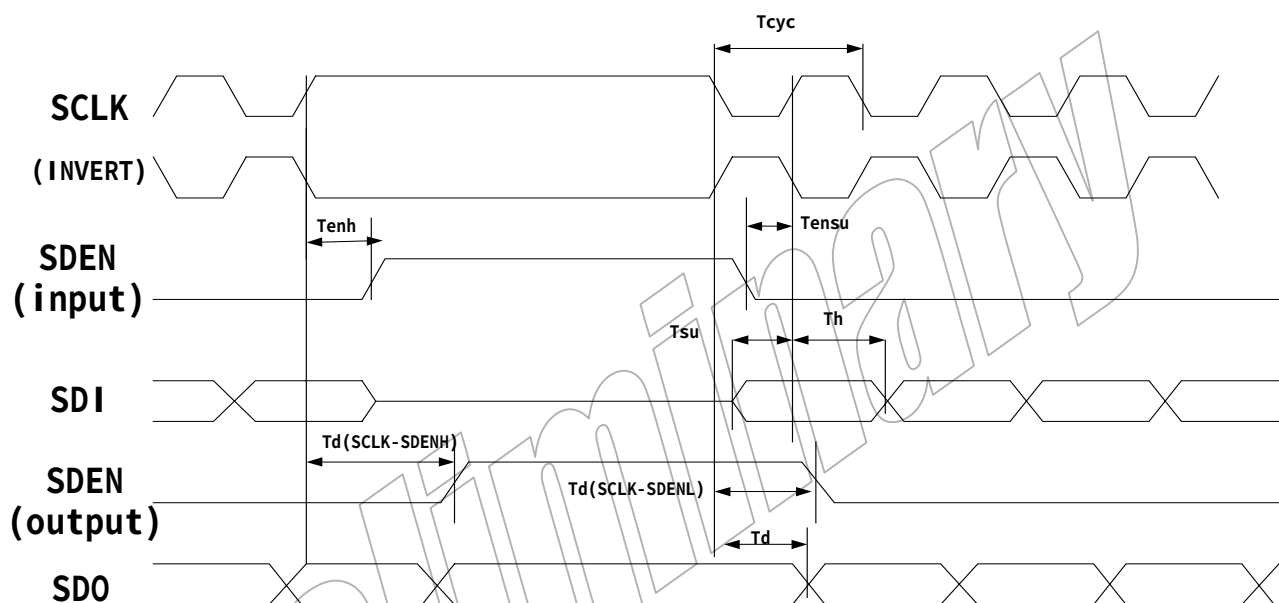


4.5 Serial Interface Timing

The case of SCLKM=0.

PARAMETER		MIN	NOM	MAX	UNIT
Tcyc	Cycle time			50	ns
Tsu	Setup time	3			ns
th	Hold time	5			ns
td	Delay time, SCLK fall(rise) to SDO			18	ns
tensu	Enable setup time before SCLK rise(fall)	3			ns
tenh	Enable hold time after SCLK rise(fall)	4			ns
Td(SCLK-SDENH)	Delay time, SCLK rise(fall) to SDEN high out			Tcyc/2+18	ns
Td(SCLK-SDENL)	Delay time, SCLK fall(rise) to SDEN low out			18	ns

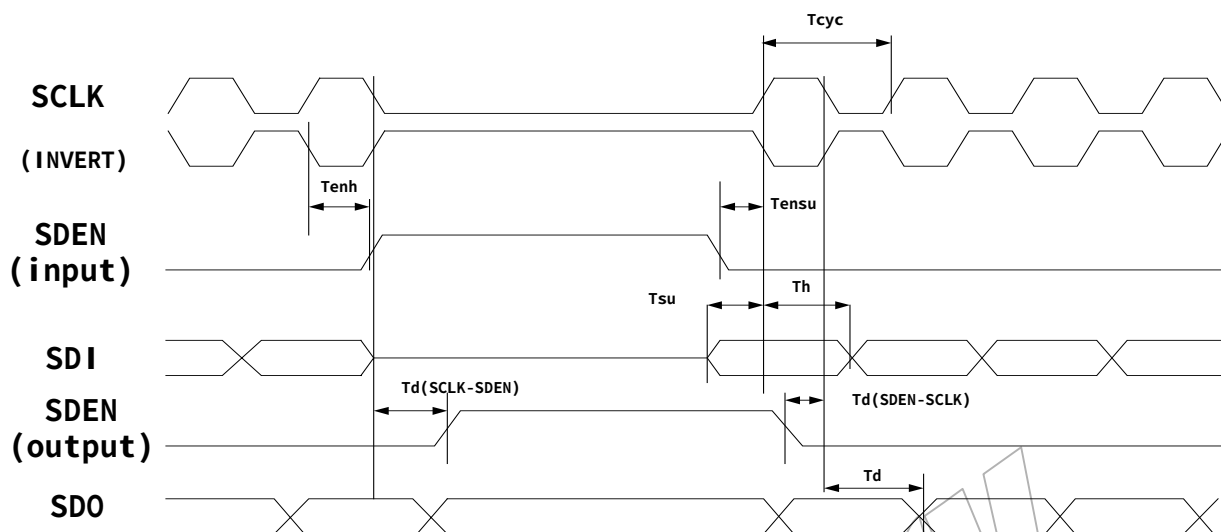
() is the case of invert clock.



The case of SCLKM=1

	PARAMETER	MIN	NOM	MAX	UNIT
Tcyc	Cycle time			25	ns
Tsu	Setup time	3			ns
Th	Hold time	5			ns
td	Delay time, SCLK fall(rise) to SDO			18	ns
tensu	Enable setup time before SCLK rise(fall)	3			ns
tenh	Enable hold time after SCLK rise(fall)	4			ns
Td(SCLK-SDEN)	Delay time, SCLK fall(rise) to SDEN high			18	ns
Td(SDEN-SCLK)	Delay time, SDEN low to SCLK fall(rise)	Tcyc-18			ns

() is the case of invert clock

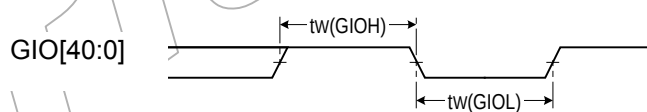


4.6 General I/O Timig

PARAMTER	MIN	NOM	MAX	UNIT
tw(GIOH) GIO[40:0] high pulse width	2Tcyc*			ns
tw(GIOL) GIO[40:0] low pulse width	2Tcyc*			ns

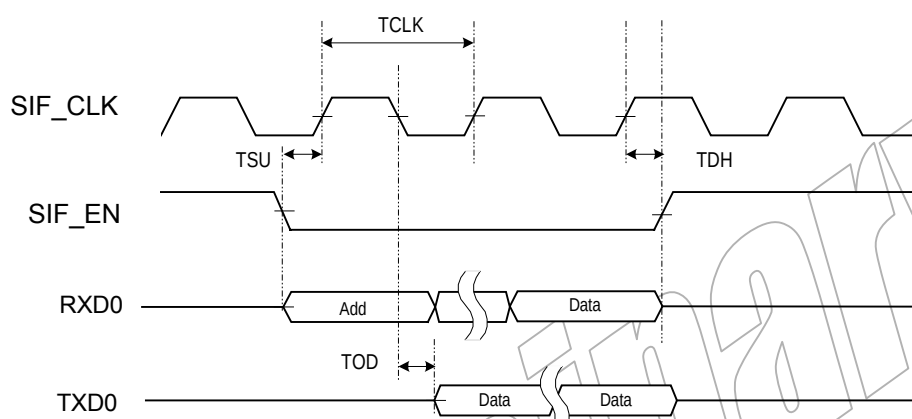
* Tcyc means one cycle time of ARM clock.

GIO external interrupt timings



4.7 AIM Serial Interface Timing

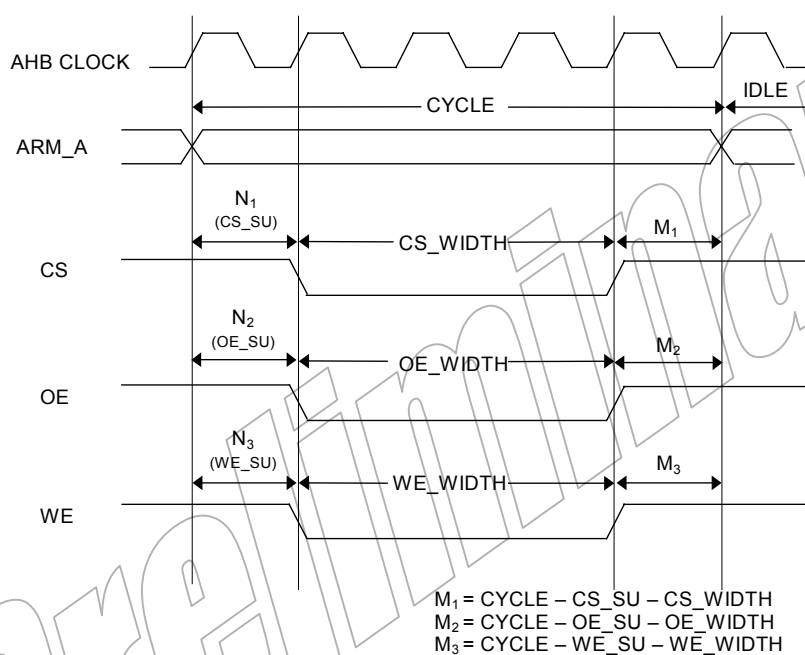
PARAMTER		MIN	NOM	MAX	UNIT
TCLK	Cycle time, (SIF_CLK)	25			ns
TSU	Input setup, (SIF_EN, RXD0)	3			ns
TDH	Input hold, (SIF_EN, RXD0)	7			ns
TOD	Output signal delay time, (TXD0)			22	ns



4.8 External Memory Interface Timing

The parameters and symbols in the following table and timing chart correspond to those in the tables and timing charts in 4.8.1 - 4.8.6.

Parameter	Description	Clock variables	Variable unit
CYCLE	Number of cycles	2-16 or 32 clocks	1 CLK (AHB_CLK)
CS_SU	Number of Setup of CS	0-15 clocks	1 CLK
OE_SU	Number of Setup of OE	"	1 CLK
WE_SU	Number of Setup of WE	"	1 CLK
CS_WIDTH	Number of Width of CS	1-16 or 32 clocks	1 CLK
OE_WIDTH	Number of Width of OE	"	1 CLK
WE_WIDTH	Number of Width of WE	"	1 CLK
IDLE	Number of Idles*	0-3 clocks	1 CLK



4.8.1 FLASH/SRAM Interface

PARAMETER	MIN	NOM	MAX	UNIT
tAD Address to CE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tBED CE to BE delay time	-2		4	ns
tDOD output data to CE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tOED Address to OE delay time	$N_2 \times T_c - 2$		$N_2 \times T_c + 4$	ns
tWED Address to WE delay time	$N_3 \times T_c - 2$		$N_3 \times T_c + 4$	ns
tDIS input data setup time to OE de-asserted	12			ns
tDIH input data hold time from OE de-asserted	0			ns
tWSU WAIT setup time to OE/WE de-asserted	$2 \times T_c + 12$			ns
tAD' CE to Address delay time	$M_1 \times T_c - 2$		$M_1 \times T_c + 4$	ns
tBED' BE to CE delay time	-2		4	ns
tDOD' CE to output data delay time	$M_1 \times T_c - 7$		$M_1 \times T_c + 4$	ns
tOED' OE to Address delay time	$M_2 \times T_c - 2$		$M_2 \times T_c + 4$	ns
tWED' WE to Address delay time	$M_3 \times T_c - 2$		$M_3 \times T_c + 4$	ns

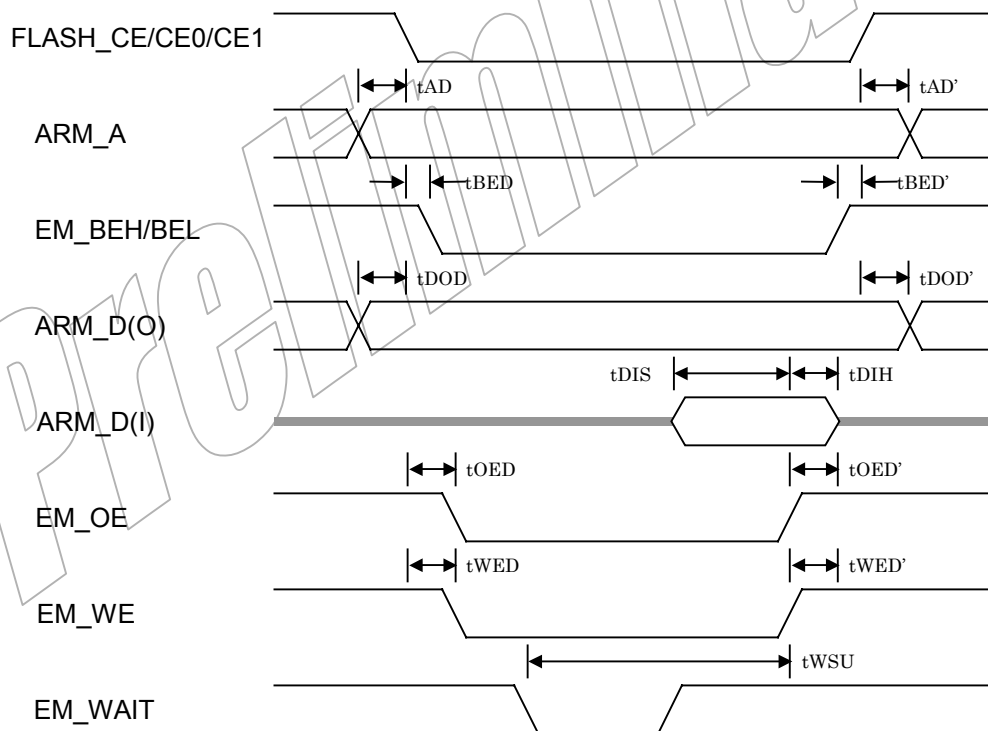
Tc: AHB clock cycle

M_1 = CYCLE - CE setup cycle - CE low cycle

M_2 = CYCLE - OE setup cycle - OE low cycle

M_3 = CYCLE - WE setup cycle - WE low cycle

FLASH&SRAM AC timing



4.8.2 COMPACT FLASH Interface

PARAMETER	MIN	NOM	MAX	UNIT
tAD Address to CE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tDOD output data to CE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tOED Address to OE delay time	$N_2 \times T_c - 2$		$N_2 \times T_c + 4$	ns
tWED Address to WE delay time	$N_3 \times T_c - 2$		$N_2 \times T_c + 4$	ns
tDIS input data setup time to OE de-asserted	12			ns
tDIH input data hold time from OE de-asserted	0			ns
tWSU WAIT setup time to OE/WE de-asserted	$2 \times T_c + 12$			ns
tAD' CE to Address delay time	$M_1 \times T_c - 2$		$M_1 \times T_c + 4$	ns
tDOD' CE to output data delay time	$M_1 \times T_c - 7$		$M_1 \times T_c + 4$	ns
tOED' OE to Address delay time	$M_2 \times T_c - 2$		$M_2 \times T_c + 4$	ns
tWED' WE to Address delay time	$M_3 \times T_c - 2$		$M_3 \times T_c + 4$	ns

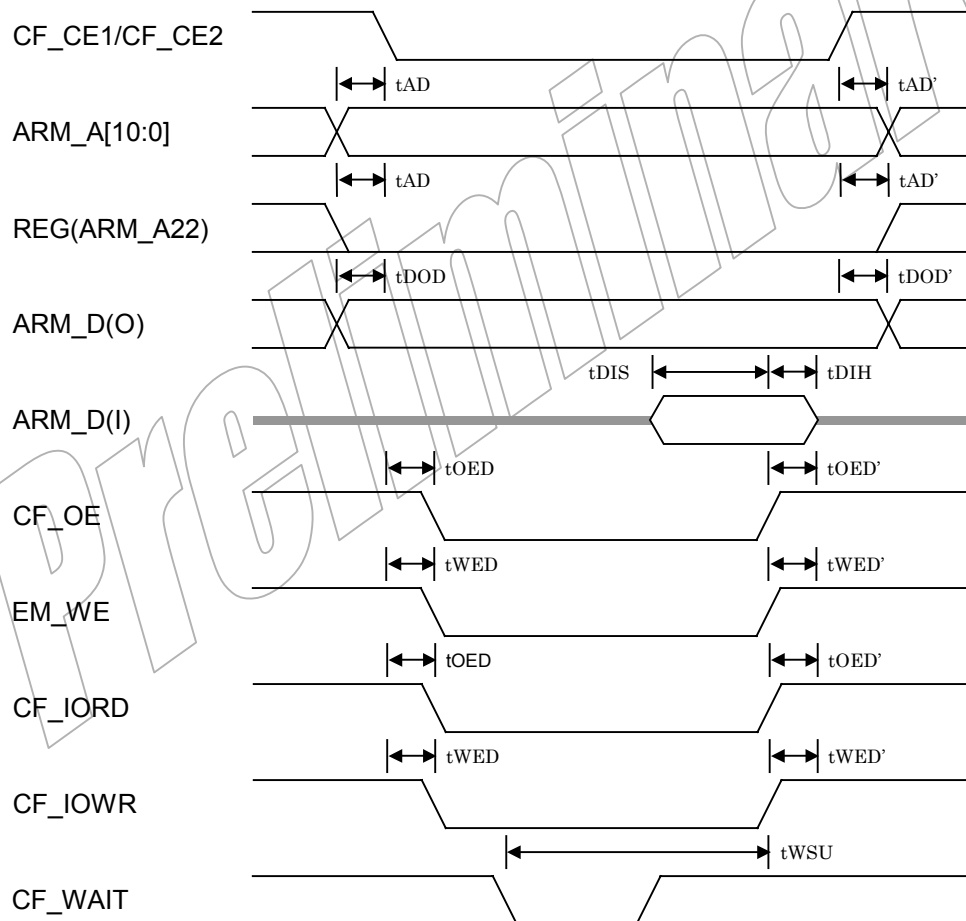
Tc: AHB clock cycle

M₁ = CYCLE - CE setup cycle - CE low cycle

M₂ = CYCLE - OE setup cycle - OE low cycle

M₃ = CYCLE - WE setup cycle - WE low cycle

CompactFlash AC timing



4.8.3 SMART MEDIA Interface

PARAMETER	MIN	NOM	MAX	UNIT
tALD ARM_D to ALE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tCLD ARM_D to CLE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tRED ARM_D to RE delay time	$N_2 \times T_c - 2$		$N_2 \times T_c + 4$	ns
tWED ARM_D to WE delay time	$N_3 \times T_c - 2$		$N_3 \times T_c + 4$	ns
tDIS input data setup time to OE de-asserted	12			ns
tDIH input data hold time from OE de-asserted	0			ns
tALD' ALE to ARM_D delay time	$M_1 \times T_c - 7$		$M_1 \times T_c + 4$	ns
tCLD' CLE to ARM_D delay time	$M_1 \times T_c - 7$		$M_1 \times T_c + 4$	ns
tRED' RE to ARM_D delay time	$M_2 \times T_c - 7$		$M_2 \times T_c + 4$	ns
tWED' ARM_D to WE delay time	$M_3 \times T_c - 7$		$M_3 \times T_c + 4$	ns

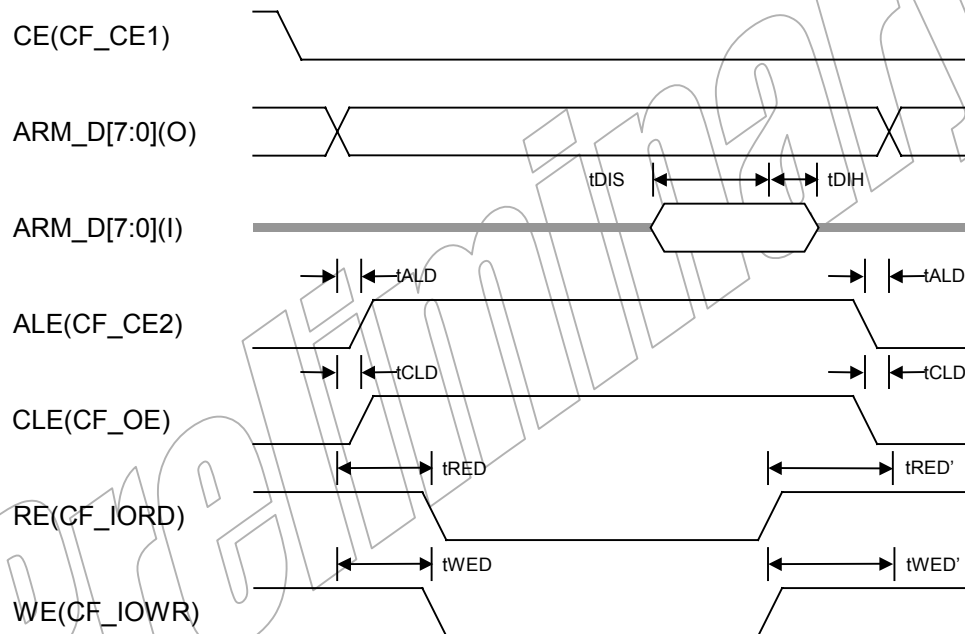
Tc: AHB clock cycle

M₁ = CYCLE - CE setup cycle - CE low cycle

M₂ = CYCLE - OE setup cycle - OE low cycle

M₃ = CYCLE - WE setup cycle - WE low cycle

SmartMedia AC timing



4.8.4 FLASH/SRAM Interface (used SDRAM bus)

PARAMETER		MIN	NOM	MAX	UNIT
tAD	Address to CE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tBED	CE to BE delay time	-2		4	ns
tDOD	output data to CE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tOED	Address to OE delay time	$N_2 \times T_c - 2$		$N_2 \times T_c + 4$	ns
tWED	Address to WE delay time	$N_3 \times T_c - 2$		$N_3 \times T_c + 4$	ns
tDIS	input data setup time to OE de-asserted	12			ns
tDIH	input data hold time from OE de-asserted	0			ns
tWSU	WAIT setup time to OE/WE de-asserted	$2 \times T_c + 12$			ns
tAD'	CE to Address delay time	$M_1 \times T_c - 2$		$M_1 \times T_c + 4$	ns
tBED'	BE to CE delay time	-2		4	ns
tDOD'	CE to output data delay time	$M_1 \times T_c - 7$		$M_1 \times T_c + 4$	ns
tOED'	OE to Address delay time	$M_2 \times T_c - 2$		$M_2 \times T_c + 4$	ns
tWED'	WE to Address delay time	$M_3 \times T_c - 2$		$M_3 \times T_c + 4$	ns

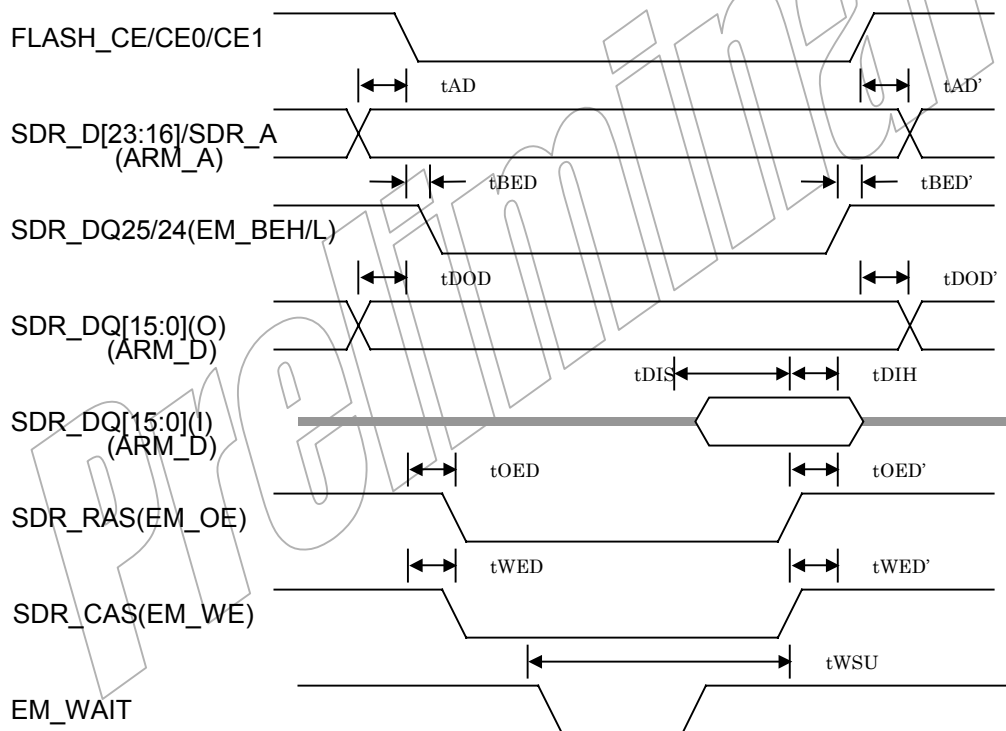
Tc: AHB clock cycle

M₁ = CYCLE - CE setup cycle - CE low cycle

M₂ = CYCLE - OE setup cycle - OE low cycle

M₃ = CYCLE - WE setup cycle- WE low cycle

FLASH&SRAM AC timing



4.8.5 COMPACT FLASH Interface (used SDRAM bus)

PARAMETER		MIN	NOM	MAX	UNIT
tAD	Address to CE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tDOD	output data to CE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tOED	Address to OE delay time	$N_2 \times T_c - 2$		$N_2 \times T_c + 4$	ns
tWED	Address to WE delay time	$N_3 \times T_c - 2$		$N_2 \times T_c + 4$	ns
tDIS	input data setup time to OE de-asserted	12			ns
tDIH	input data hold time from OE de-asserted	0			ns
tWSU	WAIT setup time to OE/WE de-asserted	$2 \times T_c + 12$			ns
tAD'	CE to Address delay time	$M_1 \times T_c - 2$		$M_1 \times T_c + 4$	ns
tDOD'	CE to output data delay time	$M_1 \times T_c - 7$		$M_1 \times T_c + 4$	ns
tOED'	OE to Address delay time	$M_2 \times T_c - 2$		$M_2 \times T_c + 4$	ns
tWED'	WE to Address delay time	$M_3 \times T_c - 2$		$M_3 \times T_c + 4$	ns

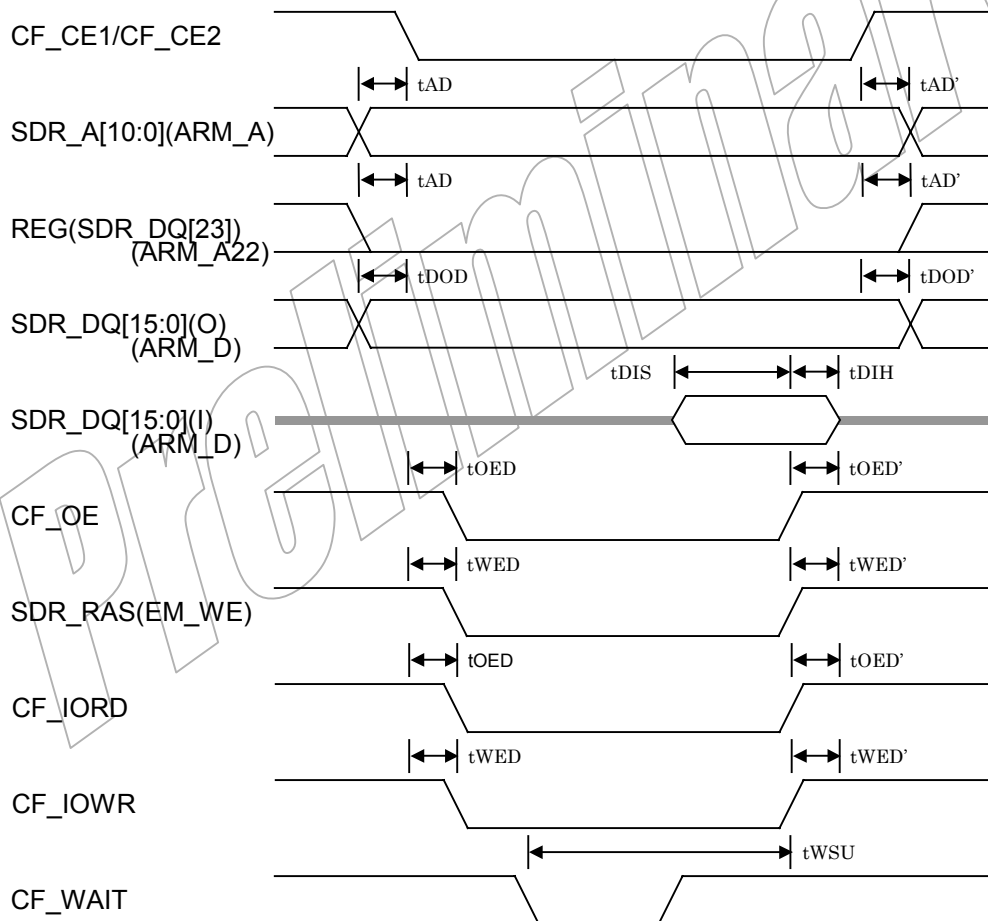
Tc: AHB clock cycle

M₁ = CYCLE - CE setup cycle - CE low cycle

M₂ = CYCLE - OE setup cycle - OE low cycle

M₃ = CYCLE - WE setup cycle - WE low cycle

CompactFlash AC timing



4.8.6 SMART MEDIA Interface (used SDRAM bus)

PARAMETER	MIN	NOM	MAX	UNIT
tALD ARM_D to ALE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tCLD ARM_D to CLE delay time	$N_1 \times T_c - 2$		$N_1 \times T_c + 4$	ns
tRED ARM_D to RE delay time	$N_2 \times T_c - 2$		$N_2 \times T_c + 4$	ns
tWED ARM_D to WE delay time	$N_3 \times T_c - 2$		$N_3 \times T_c + 4$	ns
tDIS input data setup time to OE de-asserted	12			ns
tDIH input data hold time from OE de-asserted	0			ns
tALD' ALE to ARM_D delay time	$M_1 \times T_c - 7$		$M_1 \times T_c + 4$	ns
tCLD' CLE to ARM_D delay time	$M_1 \times T_c - 7$		$M_1 \times T_c + 4$	ns
tRED' RE to ARM_D delay time	$M_2 \times T_c - 7$		$M_2 \times T_c + 4$	ns
tWED' ARM_D to WE delay time	$M_3 \times T_c - 7$		$M_3 \times T_c + 4$	ns

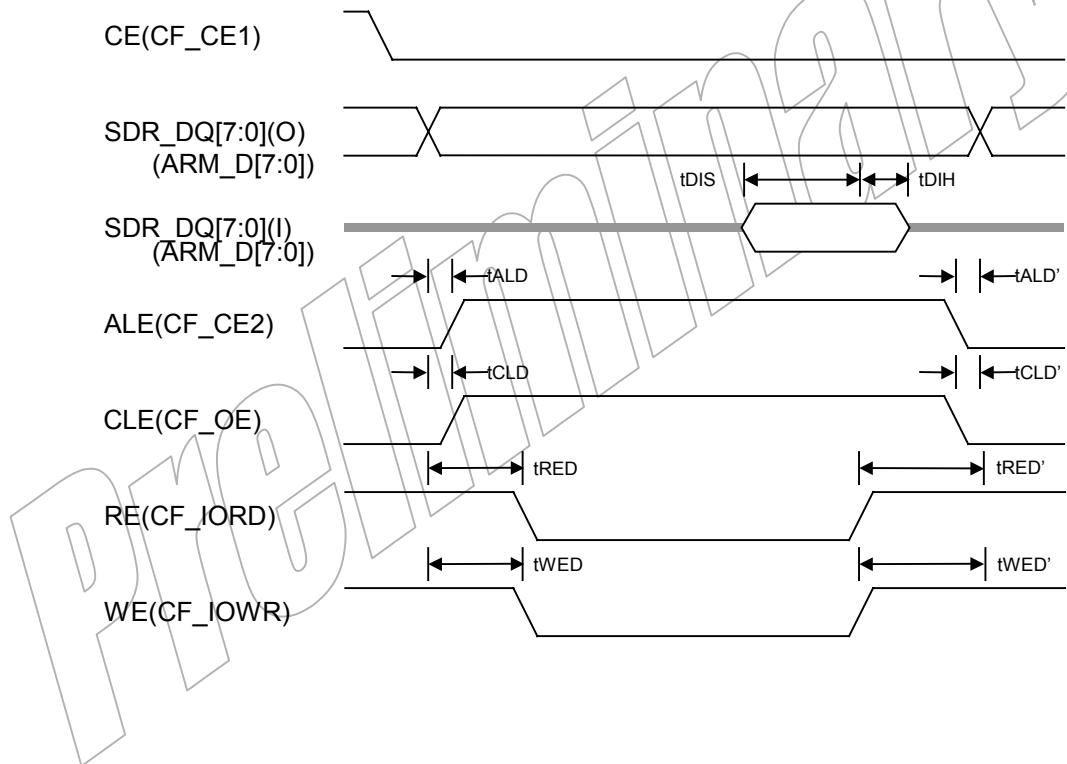
Tc: AHB clock cycle

M_1 = CYCLE - CE setup cycle - CE low cycle

M_2 = CYCLE - OE setup cycle - OE low cycle

M_3 = CYCLE - WE setup cycle - WE low cycle

SmartMedia AC timing



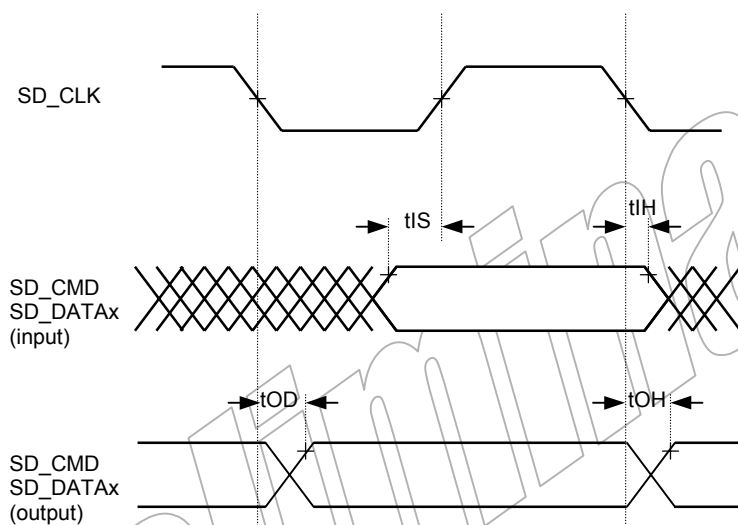
4.8.7 MultiMedia Card (MMC) Timing

Below table assumes testing over recommended operating conditions.

MultiMedia Card(MMC) Signal Switching Characteristics

	PARAMETER	MIN	NOM	MAX	UNIT
tIS	Input setup time	11			ns
tIH	Input hold time	0			ns
tOD	Output delay from SD_CLK falling			10	ns
tOH	Output hold from SD_CLK falling	0			ns

MultiMedia Card(MMC) Timings



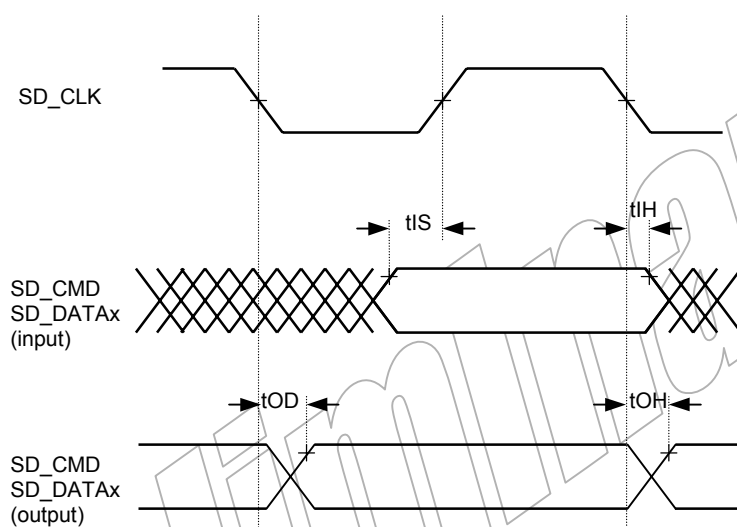
4.9 Secure Digital (SD) Timing

Below table assumes testing over recommended operating conditions.

Secure Digital(SD) Signal Switching Characteristics

PARAMETER		MIN	NOM	MAX	UNIT
tIS	Input setup time	11			ns
tIH	Input hold time	0			ns
tOD	Output delay from SD_CLK falling			10	ns
tOH	Output delay from SD_CLK rising	0			ns

Secure Digital (SD) Card Timings

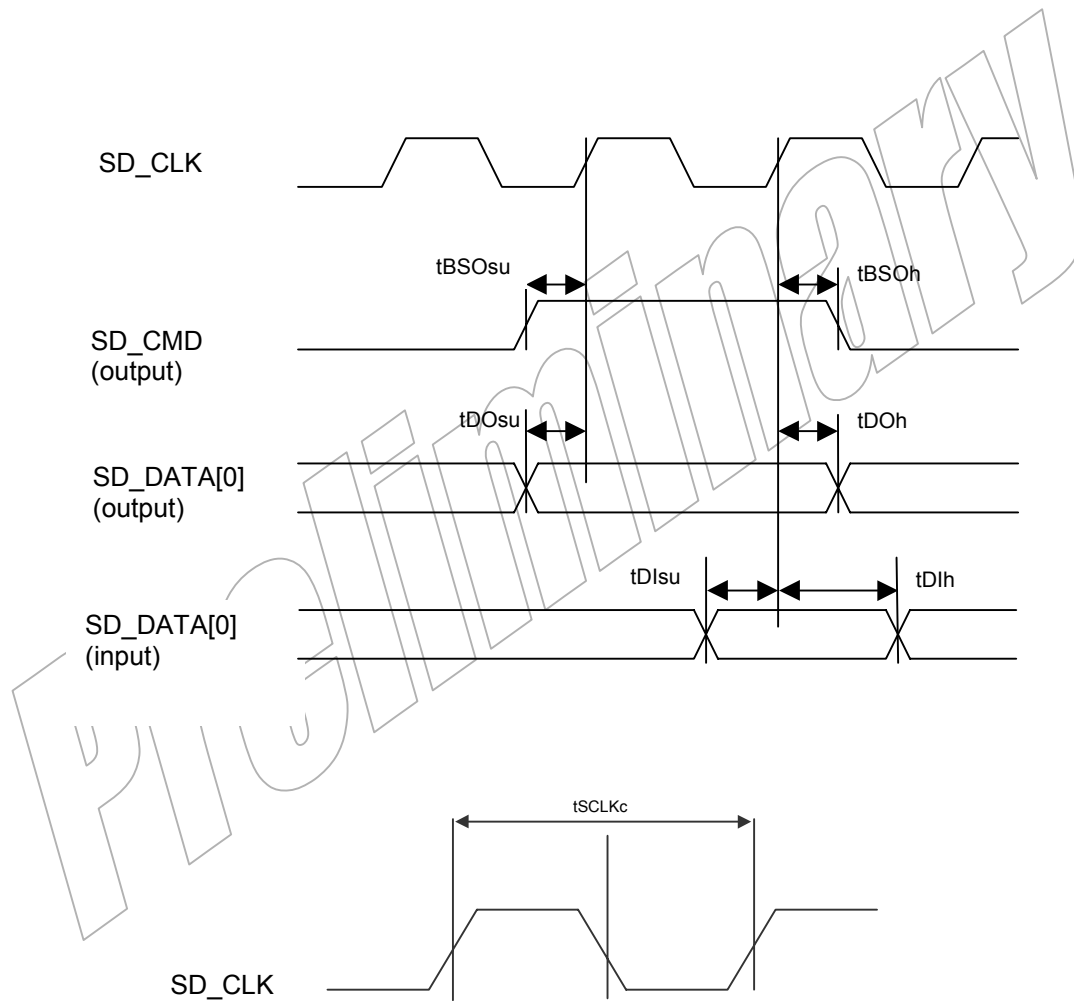


4.10 Memory Stick Timing

Memory Stick Switching Characteristics

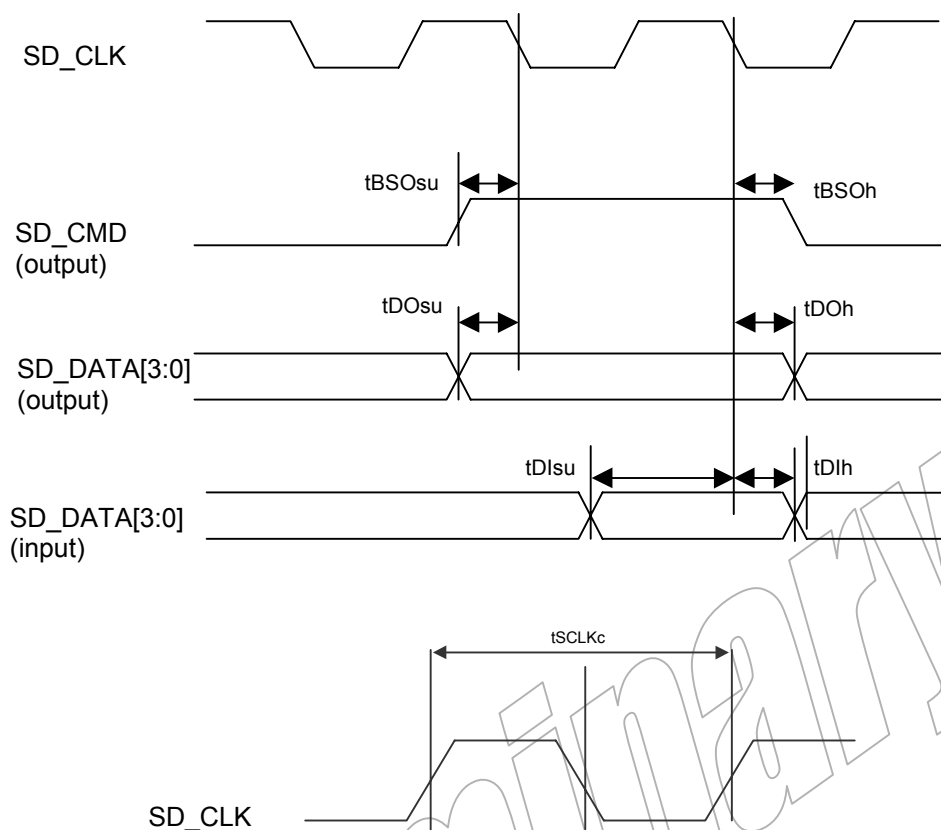
Serial Interface(MemoryStick ver1.4)

PARAMETER	MIN	MAX	UNIT
tSCLKc	Serial Clock Frequency	20	MHz
tBSOsu	Bus State output setup time	10	ns
tBSOh	Bus State output hold time	10	ns
tDISu	Data input setup time	5	ns
tDIh	Data input hold time	7	ns
tDOSu	Data output setup time	10	ns
tDOh	Data output hold time	10	ns



Parallel Interface(MemoryStick Pro)

PARAMETER	MIN	MAX	UNIT
tSCLKc	Serial Clock Frequency	40	Mhz
tBSOsu	Bus State output setup time	9	ns
tBSOh	Bus State output hold time	2	ns
tDISu	Data input setup time	6	ns
tDIh	Data input hold time	2	ns
tDOSu	Data output setup time	9	ns
tDOh	Data output hold time	2	



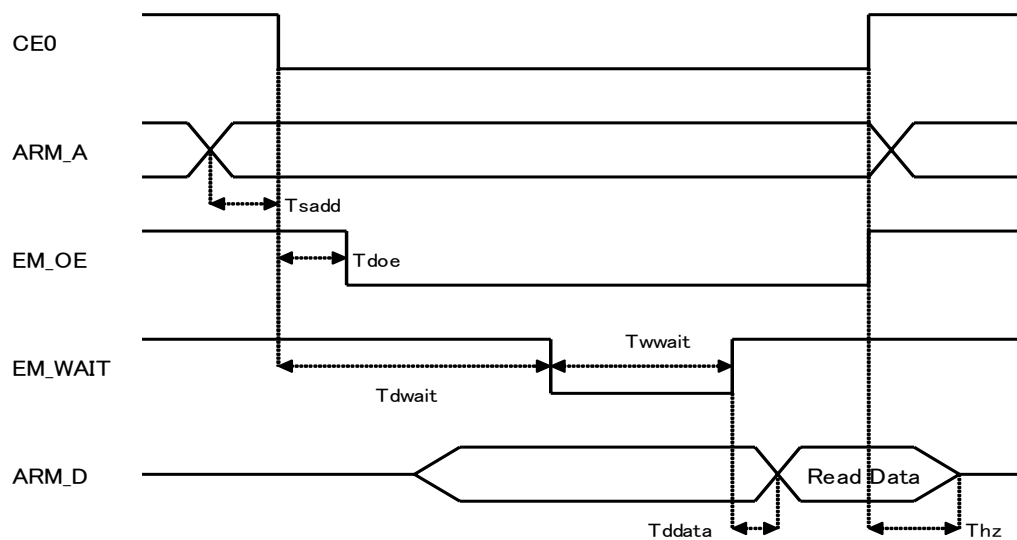
4.11 External Host Interface Timing

External Host Interface Read Timing (DM320 -> External Host)

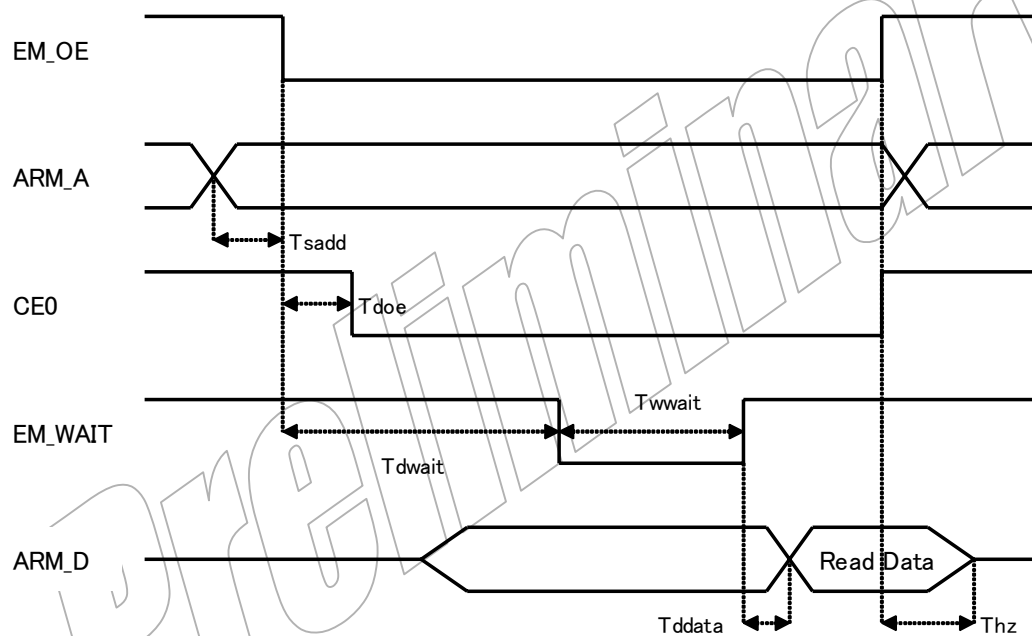
PARAMETER	MIN	NOM	MAX	UNIT
Tdoe	CE to OE input delay time	-	$T_{cyc} - 4$	ns
Tsadd	CE to Address input delay time	0		ns
Tdwait	CE to Wait output delay time	-	$3T_{cyc} + 20$	ns
Twwait	Wait width	$T_{cyc} - 2$	-	ns
Tddata	Data output delay time	-	3	ns
Thz	Data output hold time from CE de-asserted	0	12	ns
Tce0	Minimum interval time	$1T_{cyc}$	-	ns

T_{cyc} = DM320 System clock cycle period

External Host Interface Read Timing (CE trigger mode)

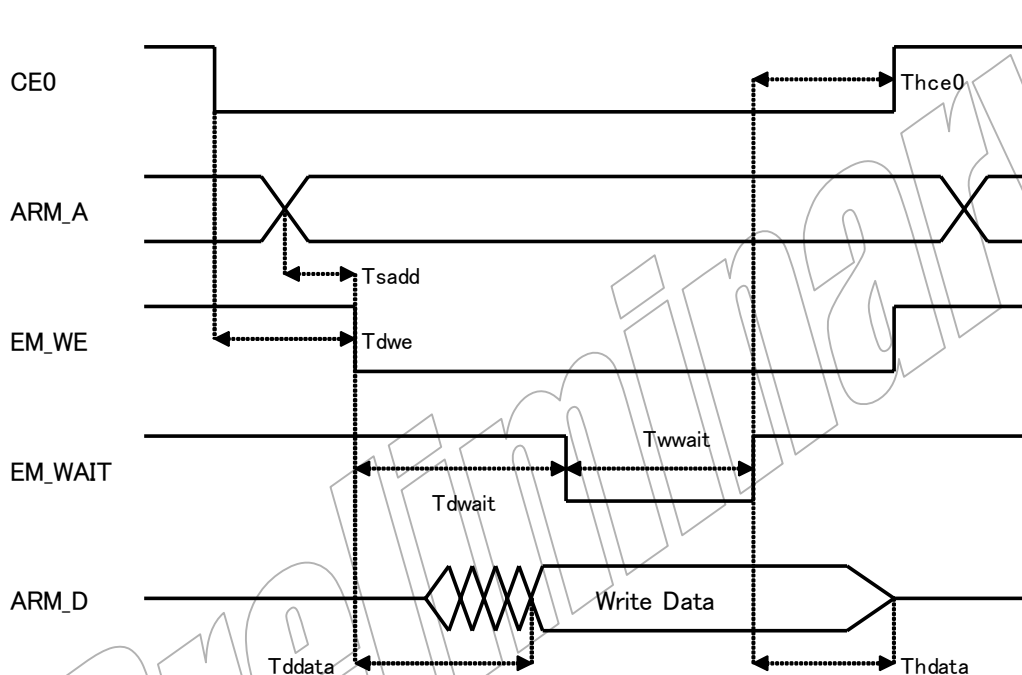


External Host Interface Read Timing (OE trigger mode)

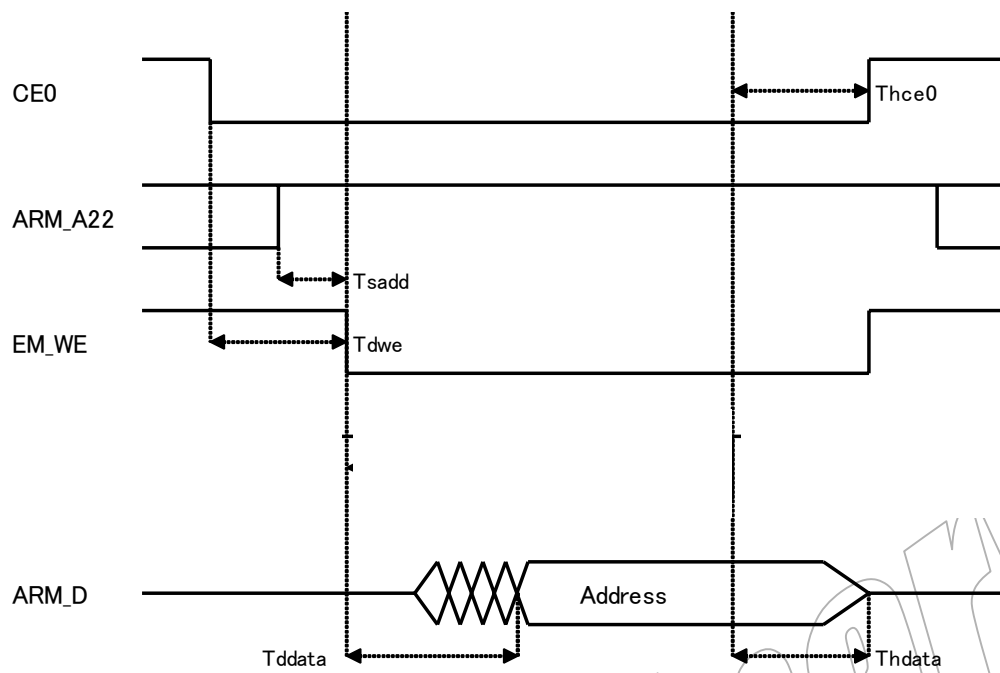


PARAMETER		MIN	NOM	MAX	UNIT
Tdwe	CE to WE input delay time	0		-	ns
Tdadd	WE to Address input delay time	0			ns
Tdwait	WE to Wait output delay time	-		3Tcyc+ 20	ns
Twwait	Wait width	Tcyc - 2		-	ns
Tddata	WE to data input delay time	-		Tcyc – 4	ns
Thce0	CE hold time from Wait de-asserted	0		-	ns
Thdata	Data hold time from Wait de-asserted	0		-	ns
Tce0	Minimum interval time	1Tcyc		-	ns

External Host Interface Write Timing

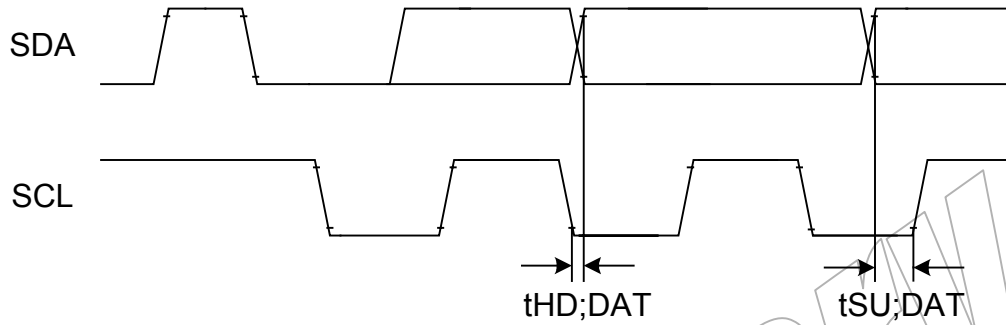


PARAMETER		MIN	NOM	MAX	UNIT
Tdwe	CE to WE input delay time	0		-	ns
Tsbmux	EH_BMUX to WE input delay time	0			
Tsadd	WE to Address input delay time	0			ns
Tdwait	WE to Wait output delay time	-		Tcycx2 + 20	ns
Twwait	Wait width	Tcyc - 2		-	ns
Tddata	WE to data input delay time	-		Tcyc - 4	ns
Thce0	CE hold time from Wait de-asserted	0		-	ns
Thdata	Data hold time from Wait de-asserted	0		-	ns
Thbmux	EH_BMUX hold time to EM_WAIT	0			
Tce0	Minimum interval time	1Tcyc		-	ns



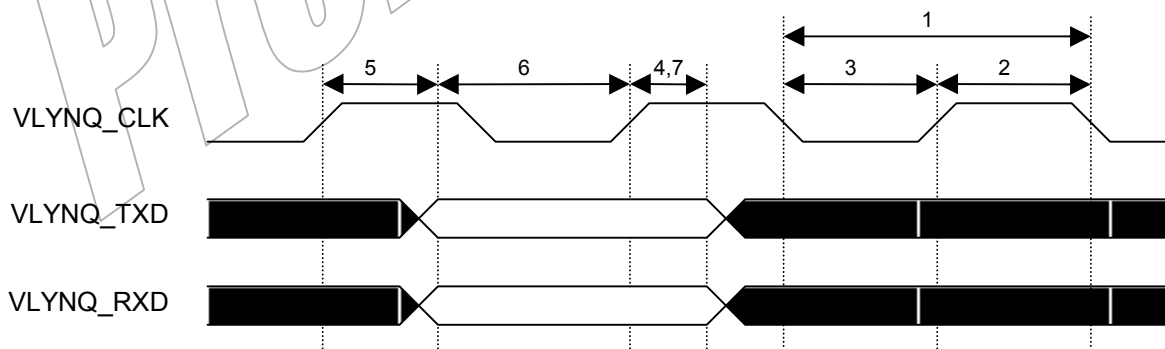
4.12 Pseudo I²C Bus Timing

PARAMETER	MIN	NOM	MAX	UNIT
t _{HD:DAT} Data hold (or delay) time(400kHz)			700	ns
t _{HD:DAT} Data hold (or delay) time(100kHz)			5100	ns
t _{SU:DAT} Data set-up time	0		-	ns



4.13 VLYNQ

No.	Parameter	Min	Max	Unit
1	Clock period (T _{period}), VLYNQ_CLK	9.26		ns
2	Pulse duration, VLYNQ_CLK high	t _{bd}		ns
3	Pulse duration, VLYNQ_CLK low	t _{bd}		ns
4	Delay time, VLYNQ_CLK to VLYNQ_TXD invalid	2.25		ns
5	Delay time, VLYNQ_CLK to VLYNQ_TXD valid		9.0	ns
6	Setup time, VLYNQ_RXD before VLYNQ_CLK	0		ns
7	Hold time, VLYNQ_RXD after VLYNQ_CLK	2.0		ns



5. DAC

- DESCRIPTION
 - 10bit D/A Converter

Signal Name and Function

SIGNAL NAME (*1)	I/O	FUNCTION
BIAS	IN	Bias with phase compensation capacitor
IREF	IN	Current reference control
VREF	IN	Reference voltage
AOUT	OUT	Analog output

Recommended Operating Conditions

PARAMETER	MIN	TYP	MAX	UNIT
FCLK			27	MHz
VO (*1)	VSSA		Vref	V
Vref	1.0		1.3	V
Riref		1.6		kΩ
Raout		200		Ω
Cbias		0.1		μF

*1) This VO range is kept while the ratio (Riref:Raout) have 8:1 relationship.

Operating Characteristics over Recommended Operating Free-Air Temperature

(V_{DD}=1.5V, V_{DD3V}=3.3V, V_{DDA}=3.3V, V_{ref}=1V, R_{aout}=200Ω, R_{ref}=1.6KΩ)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
ZSET	Zero-scale offset			10	LSB
FSET	Full-scale offset			10	LSB
SINL	Static integral non-linearity Error	f _{clk} =27MHz		±1.0	LSB
SDNL	Static differential non-linearity Error	f _{clk} =27MHz		±0.5	LSB

6. Power Supply Sequencing

It is better that core logic states be stable during power-up/power-down transitions for DVDD/DVDD_SDR/USBVDD/DAC power supply. The internal states of inputs and outputs are undefined during the CVDD power-up/power-down transitions and outputs can range from high impedance to full rated source or sink current. Some of the possible states could create bus-contention conditions if other devices are driving the bus. For these reasons, it is preferred design practice to bring up CVDD/PLLVDD first and DVDD/DVDD_SDR/USBVDD/DAC power supply last(or DVDD/DVDD_SDR second and USBVDD/DAC power supply last) ,and bring down USBVDD/DAC/DVDD/DVDD_SDR first(or USBVDD/DAC power supply first, DVDD/DVDD_SDR second), and CVDD/PLLVDD down last.

When there is no probability of bus-contention conditions, the recommended operating conditions listed in the Common Characteristics section for the specific product are adequate to protect the device.

7. Package

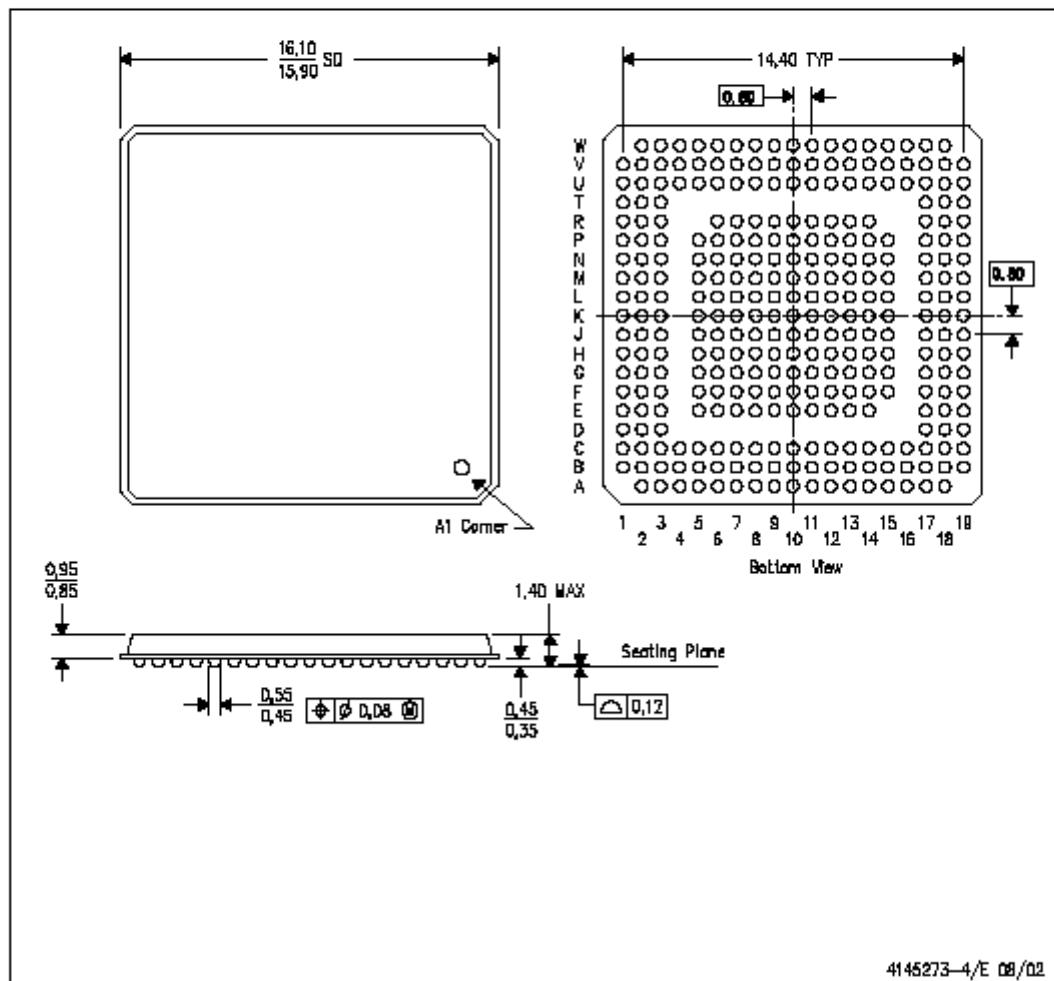
7.1 306ZHK Size Table

Dimension	W	16.00+/-0.10mm
	D	16.00+/-0.10mm
	H	1.4mm max
Ball Pitch		0.8mm

7.2 306ZHK Drawing

GHK (S-PBGA-N306)

PLASTIC BALL GRID ARRAY



NOTES:

- All linear dimensions are in millimeters.
- This drawing is subject to change without notice.
- Micro Star BGA configuration.

SCALE	SIZE	4145273	REV	SHEET
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